Likelihood-ratio tests for continuous outcomes truncated by death

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Abstract

Continuous outcomes such as quality-of-life scores are often undefined after death or a related event. A common pragmatic solution is to encode such patients by the worst possible score and analyze the resulting composite outcome with a one-dimensional procedure. That preserves a clinically meaningful ordering, but it conflates treatment effects on the atom with treatment effects on the continuous part among those with observed outcomes. We propose likelihood-ratio tests for the joint null of no treatment effect on either component. The construction yields a single prespecified p-value together with interpretable component-wise effect estimates and confidence regions. We present both a parametric version and a semi-parametric version based on empirical likelihood, establish their large-sample calibration, and study their behavior in simulations. The methods are most advantageous when treatment acts primarily through one component or when the atom and continuous components move in different directions. A clinical application to the IST-3 acute ischemic stroke trial illustrates the method using death by six months as the atom and EuroQol visual analogue score among survivors as the continuous component.

Keywords: Quality of Life, Randomized Controlled Trials,
Truncated Outcomes, Semi-Continuous Distributions, Empirical Likelihood

1 Introduction

Continuous outcomes such as quality-of-life scores, cognition scores, and days alive outside hospital are increasingly used in randomized trials of serious illness. When patients die before assessment, however, those outcomes are undefined rather than merely missing, creating the truncation-by-death problem (Frangakis and Rubin, 2002; Colantuoni et al., 2018). Composite outcomes that combine mortality with an otherwise continuous scale are common in critical care and related areas; familiar examples include ventilator-free days and days alive without life support (The COVID STEROID 2 Trial Group, 2021; Yehya et al., 2019).

One pragmatic response is to place death, or another substantively distinct event, at the bottom of a single ordered scale and analyze the resulting composite with a rank-based or related one-dimensional method. Worst-rank analyses and related composite rankings have been advocated precisely because they retain all randomized patients in the primary analysis (Lachin, 1999; McMahon and Harrell, 2000; Colantuoni et al., 2018). The same logic underlies the widespread use of atom-at-zero outcomes such as ventilator-free days, where death is assigned the worst possible

value on the scale (Yehya et al., 2019). Throughout, we refer to the Mann–Whitney–Wilcoxon rank-sum test simply as the Wilcoxon test (Wilcoxon, 1945; Mann and Whitney, 1947).

That strategy is often clinically sensible, but it targets the distribution of the composite outcome itself. It does not separate treatment effects on the probability of the atom from treatment effects on the continuous outcome among patients with observed values. When the two components move in different directions, or when treatment acts mainly through one of them, a one-dimensional analysis can be attenuated because opposite component effects partially cancel on the combined scale. The issue is therefore not only one of robustness or convenience; it is also one of estimand alignment.

Other approaches address truncated-by-death outcomes from different perspectives. Survivor-only analyses condition on a post-randomization event and therefore answer a conditional question that can differ materially from the intention-to-treat comparison. Principal-stratification and survivor-average-causal-effect approaches instead target causal effects in latent survivor strata, but they require assumptions that are not identified from the observed trial data alone (Frangakis and Rubin, 2002; Colantuoni et al., 2018). Composite-endpoint approaches avoid that latent-stratum target by working directly with an ordered endpoint, but then the estimand is tied to the chosen ranking (McMahon and Harrell, 2000; Colantuoni et al., 2018). Our goal in this paper is narrower and more operational: we target the joint observed-data null of no treatment effect on the atom probability and no treatment effect on the continuous component among those for whom it is observed.

Our construction combines ideas from two-part models for semi-continuous outcomes (Aitchison, 1955; Cragg, 1971) and empirical-likelihood inference for mean restrictions (Owen, 1988, 1991; Kitamura, 2001). We develop a factorized likelihood-ratio framework that yields a single prespecified p-value for the joint null together with interpretable component-wise effect estimates and confidence regions. We present both a parametric version, which is efficient under a credible model for the positive part, and a semi-parametric version, which relaxes that distributional commitment while retaining the same Bernoulli component. The resulting procedures preserve the clinically meaningful atom used in practice, but they test a sharper question than direct one-dimensional analyses of the combined outcome.

Although we motivate the method by quality of life truncated by death, the same structure arises whenever one special value has a substantively different meaning than the remainder of an otherwise continuous scale. The target is an observed-data treatment contrast, not a principal-stratum causal estimand. The rest of the paper is organized as follows. Section 2 introduces the statistical setup and the proposed tests. Section 3 reports a simulation study, Section 4 gives a clinical application, and Section 5 concludes. Additional simulation results and a derivation of the empirical-likelihood component are collected in Appendix A and Appendix C.

2 Method

2.1 Setup and scientific null

We observe independent and identically distributed random variables (Y_i, A_i, R_i, X_i) for $i = 1, \ldots, n$. Here Y_i is a continuous outcome when it is defined, $A_i \in \{0, 1\}$ indicates whether the outcome is defined and observed, $R_i \in \{0, 1\}$ is the treatment indicator, and X_i is a p-dimensional vector of

baseline covariates. Because Y_i is only meaningful when $A_i = 1$, the scientific null hypothesis of no treatment effect is

$$\mathcal{L}(Y_i \mid A_i = 1, R_i = 1, X_i) = \mathcal{L}(Y_i \mid A_i = 1, R_i = 0, X_i),$$
$$\Pr(A_i = 1 \mid R_i = 1, X_i) = \Pr(A_i = 1 \mid R_i = 0, X_i).$$

In practice, analysts often replace this bivariate object by a single combined outcome. Let $\mathcal{Y} \subset \mathbb{R}$ be a compact set containing the support of $Y_i \mid A_i = 1$, and let $\mathcal{E} \notin \mathcal{Y}$ be a distinguished symbol representing the undefined outcome. We define the outcome space by

$$\widetilde{\mathcal{Y}} = \mathcal{Y} \sqcup \{\mathcal{E}\},$$

where $\sqcup$ denotes disjoint union. We equip $\widetilde{\mathcal{Y}}$ with the σ-algebra

$$\widetilde{\mathcal{B}} = \{B,\ B \cup \{\mathcal{E}\} : B \in \mathcal{B}(\mathcal{Y})\},$$

where $\mathcal{B}(\mathcal{Y})$ is the Borel σ-algebra on $\mathcal{Y}$ induced by $\mathbb{R}$. Thus $\mathcal{E}$ is an abstract atom, distinct from every point of $\mathcal{Y}$, and $\widetilde{Y}_i$ is a $\widetilde{\mathcal{Y}}$-valued random variable. For fixed $e_0 \in \mathbb{R}$, we define the numerical representation map

$$c_{e_0} : \widetilde{\mathcal{Y}} \to \mathbb{R}, \qquad c_{e_0}(y) = y \text{ for } y \in \mathcal{Y}, \qquad c_{e_0}(\mathcal{E}) = e_0,$$

and arithmetic is performed on $c_{e_0}(\widetilde{Y}_i)$, not on the abstract atom itself. When no ambiguity arises, one may equivalently work directly on the numerical outcome space $\mathcal{Y} \cup \{e_0\} \subseteq \mathbb{R}$, thereby identifying the atom with the actual number e_0. Throughout most of the manuscript we use the atom-at-zero representation c_0, obtained by setting $e_0 = 0$.

$$\widetilde{Y}_i = \begin{cases} Y_i, & A_i = 1, \\ \mathcal{E}, & A_i = 0. \end{cases} \tag{1}$$

where $\mathcal{E}$ is a fixed atom representing the undefined outcome. The distribution of $\widetilde{Y}_i$ is therefore a mixture of a point mass at $\mathcal{E}$ and a continuous distribution on the support of $Y_i \mid A_i = 1$.

$$\widetilde{Y}_i^{\,0} := c_0(\widetilde{Y}_i) = \begin{cases} Y_i, & A_i = 1, \\ 0, & A_i = 0. \end{cases}$$

This is the real-valued combined outcome used whenever expectations or other arithmetic summaries are taken.

The conditional mean of the numerical combined outcome is

$$\mathrm{E}[\widetilde{Y}_i^{\,0} \mid R_i, X_i] = \mathrm{E}[Y_i \mid A_i = 1, R_i, X_i] \Pr(A_i = 1 \mid R_i, X_i).$$

This quantity can be identical between treatment groups even when treatment affects both the observation probability and the continuous component. To see this, let

$$\mathrm{E}[Y_i \mid A_i = 1, R_i = 0, X_i] = \mu(X_i), \qquad \mathrm{E}[Y_i \mid A_i = 1, R_i = 1, X_i] = \mu(X_i) + \mu_\delta(X_i),$$

Then

$$\Delta_{\widetilde{Y}}(X_i) = \mathrm{E}[\widetilde{Y}_i^{\,0} \mid R_i = 1, X_i] - \mathrm{E}[\widetilde{Y}_i^{\,0} \mid R_i = 0, X_i] \tag{2}$$

$$
\begin{aligned}
&= \mu(X_i)\left[\Pr(A_i = 1 \mid R_i = 1, X_i) - \Pr(A_i = 1 \mid R_i = 0, X_i)\right] \\
&\quad + \mu_\delta(X_i)\Pr(A_i = 1 \mid R_i = 1, X_i).
\end{aligned}
$$

Here $\Delta_{\widetilde{Y}}(X_i)$ is a contrast on the atom-at-zero numerical representation $\widetilde{Y}_i^{\,0}$, not a mean on the abstract disjoint-union space. Choosing

$$
\mu_\delta(X_i) = \mu(X_i)\left[\Pr(A_i = 1 \mid R_i = 0, X_i)\Pr(A_i = 1 \mid R_i = 1, X_i)^{-1} - 1\right]
$$

forces $\Delta_{\widetilde{Y}}(X_i) = 0$ even though both components depend on treatment. Thus testing $\Delta_{\widetilde{Y}}(X_i) = 0$ is generally not equivalent to testing no treatment effect on the joint observed-data structure, and the latter requires two restrictions.

In the abstract formulation, $\mathcal{E}$ is a fixed prespecified atom representing the clinically worst state on the combined scale. When numerical summaries are required, we pass to a real-valued representation c_{e_0}; throughout most of the manuscript, the default choice is $e_0 = 0$. In applications where the observed endpoint truly has an atom at a real number such as 0, we may work directly on that numerical scale. The event $A_i = 0$ therefore denotes the substantive atom event, such as death before assessment or zero days alive out of hospital, rather than ordinary measurement missingness. Our estimands are defined with respect to the observed-data law of $(A_i, Y_i \mid A_i = 1)$, so identifiability is with respect to that observed-data structure rather than a latent principal stratum. The inferential development below assumes positive probabilities of $A_i = 1$ in both treatment groups, identifiable finite-dimensional models for the Bernoulli component, and, when the parametric procedure is used, a correctly specified model for the conditional outcome distribution.

To exploit that structure, we model the binary and continuous components separately. Let

$$
\begin{aligned}
A_i \mid R_i, X_i &\sim \text{Bernoulli}\left(\pi_i\right), \\
Y_i \mid A_i = 1, R_i, X_i &\sim F\left(\cdot; \mu_i, \Psi\right),
\end{aligned}
$$

where $\pi_i = \Pr(A_i = 1 \mid R_i, X_i)$ is the observation probability, $\mu_i = \mathrm{E}[Y_i \mid A_i = 1, R_i, X_i]$ is a mean parameter for the observed outcomes, and Ψ collects nuisance parameters governing the distribution of the continuous component. If $f(\cdot; \mu_i, \Psi)$ denotes the corresponding density, the observed-data likelihood is

$$
L_n(\mu_i, \pi_i, \Psi) = \prod_{i=1}^{n} f\left(Y_i; \mu_i, \Psi\right)^{A_i} \pi_i^{A_i}\left(1 - \pi_i\right)^{1-A_i}.
$$

This factorizes as

$$
L_n(\mu_i, \pi_i, \Psi) = L_{Y,n}(\mu_i, \Psi) L_{A,n}(\pi_i),
$$

where

$$
\begin{aligned}
L_{Y,n}(\mu_i, \Psi) &= \prod_{i=1}^{n} f\left(Y_i; \mu_i, \Psi\right)^{A_i}, \\
L_{A,n}(\pi_i) &= \prod_{i=1}^{n} \pi_i^{A_i}\left(1 - \pi_i\right)^{1-A_i}.
\end{aligned}
$$

The factorization follows from the likelihood construction; it does not require independence between the value of Y_i and the event $A_i = 1$.

We now impose finite-dimensional regression models on the two components. Let $s_\beta(x) = h_\beta(x)^\top \eta_\beta$ and $s_\mu(x) = h_\mu(x)^\top \eta_\mu$, where h_β and h_μ are prespecified transformations of the baseline covariates and η_β, η_μ are nuisance parameters. The working models used for covariate adjustment are

$$\operatorname{logit} \Pr(A_i = 1 \mid R_i, X_i) = \beta_0 + \beta_\delta R_i + s_\beta(X_i), \tag{3}$$

$$\mathrm{E}[Y_i \mid A_i = 1, R_i, X_i] = \mu_0 + \mu_\delta R_i + s_\mu(X_i), \tag{4}$$

so that treatment enters additively on the logit scale for the observation probability and additively on the mean scale for the continuous non-atom outcome. The parameter μ_δ is the conditional mean difference among observed outcomes at fixed covariates, and β_δ is the conditional log odds-ratio of being observed. We write $\alpha_\delta = \exp(\beta_\delta)$ for the corresponding odds ratio.

On the original outcome scale, μ_δ quantifies how treatment shifts the positive part of the outcome distribution among subjects with observed outcomes, whereas β_δ quantifies how treatment changes the odds that the outcome is observed at all. Reporting the pair therefore separates improvement in the continuous component from improvement in the probability of escaping the atom. This two-component contrast is the scientific target of the proposed likelihood-ratio test. It remains informative even when a single combined-outcome contrast such as $\Delta_{\widetilde{Y}}(X_i)$ is zero because the two components offset each other.

For fixed treatment contrasts $(\mu_\delta, \beta_\delta)$, the profile likelihood-ratio statistic is

$$\begin{aligned} W_n^P(\mu_\delta, \beta_\delta) &= W_{Y,n}^P(\mu_\delta) + W_{A,n}^P(\beta_\delta), \\ W_{Y,n}^P(\mu_\delta) &= -2 \log \frac{\sup\limits_{\mu_0, s_\mu, \Psi} L_{Y,n}(\mu_0, \mu_\delta, s_\mu, \Psi)}{\sup\limits_{\mu_0, \mu_\delta, s_\mu, \Psi} L_{Y,n}(\mu_0, \mu_\delta, s_\mu, \Psi)}, \\ W_{A,n}^P(\beta_\delta) &= -2 \log \frac{\sup\limits_{\beta_0, s_\beta} L_{A,n}(\beta_0, \beta_\delta, s_\beta)}{\sup\limits_{\beta_0, \beta_\delta, s_\beta} L_{A,n}(\beta_0, \beta_\delta, s_\beta)}. \end{aligned} \tag{5}$$

The null hypothesis of no treatment effect is tested with $W_n^P(0, 0)$.

2.2 Parametric likelihood-ratio test

In the parametric version, the continuous non-atom outcome distribution is specified fully. In the Gaussian working model considered here,

$$\begin{aligned} A_i \mid R_i = r, X_i = x &\sim \text{Bernoulli}\{\pi_r(x)\}, \\ Y_i \mid A_i = 1, R_i = r, X_i = x &\sim N\{\mu_r(x), \sigma^2\}, \end{aligned} \tag{6}$$

where

$$\pi_r(x) = \overset{-1}{\operatorname{logit}}\{\beta_0 + \beta_\delta r + s_\beta(x)\}, \qquad \mu_r(x) = \mu_0 + \mu_\delta r + s_\mu(x), \qquad r \in \{0, 1\}.$$

The residual variance σ^2 is common to the two treatment arms, conditional on the adjustment covariates. Let $\theta_P = (\beta_0, \beta_\delta, \eta_\beta^\top, \mu_0, \mu_\delta, \eta_\mu^\top, \sigma^2)^\top$ collect all finite-dimensional parameters. The observed-data likelihood under the Gaussian model is

$$L_n^P(\theta_P) = \prod_{i=1}^n \left[\pi_{R_i}(X_i)^{A_i} \{1 - \pi_{R_i}(X_i)\}^{1-A_i} \right]$$

$$\times \prod_{i=1}^{n} \left[(2\pi\sigma^2)^{-1/2} \exp\left\{ -\frac{\{Y_i - \mu_{R_i}(X_i)\}^2}{2\sigma^2} \right\} \right]^{A_i}. \tag{7}$$

Thus the likelihood is the product of a Bernoulli likelihood for whether the outcome escapes the atom and a Normal likelihood for the observed continuous values. The null model for the joint test imposes $\mu_\delta = 0$ and $\beta_\delta = 0$; the remaining regression parameters and σ^2 are nuisance parameters profiled out in Equation (5).

The regression formulation also induces marginal summaries standardized to a common covariate distribution. Let $\mathbb{P}_n$ denote the empirical distribution of the observed baseline covariates, so that $\mathbb{P}_n g(X) = n^{-1} \sum_{i=1}^{n} g(X_i)$. For $r \in \{0, 1\}$ define

$$\bar{\pi}_{r,n} = \mathbb{P}_n\{\pi_r(X)\}, \qquad \bar{\rho}_{r,n} = 1 - \bar{\pi}_{r,n}, \qquad \bar{\mu}_{r,n} = \mathbb{P}_n\{\mu_r(X)\}. \tag{8}$$

Here $\bar{\pi}_{r,n}$ is the observation probability that would be obtained if all subjects were assigned treatment arm r while retaining the empirical covariate distribution, $\bar{\rho}_{r,n}$ is the corresponding atom probability, and $\bar{\mu}_{r,n}$ is the standardized mean of the continuous outcome among non-atom observations. The component contrasts are

$$\delta_{\pi,n} = \bar{\pi}_{1,n} - \bar{\pi}_{0,n}, \qquad \delta_{\rho,n} = \bar{\rho}_{1,n} - \bar{\rho}_{0,n} = -\delta_{\pi,n}, \qquad \delta_{\mu,n} = \bar{\mu}_{1,n} - \bar{\mu}_{0,n}.$$

Under the additive mean model in Equation (4), $\delta_{\mu,n} = \mu_\delta$ for every empirical covariate distribution. By contrast, the observation-probability contrast $\delta_{\pi,n}$ is a marginal standardization of the conditional log-odds model, whereas β_δ and $\alpha_\delta = \exp(\beta_\delta)$ remain the conditional log odds-ratio and odds ratio, respectively.

For a real-valued representation c_{e_0} of the atom, the covariate-standardized combined-outcome contrast induced by the parametric model is

$$\Delta_n^{(e_0)} = \mathbb{P}_n\Big[e_0\{1 - \pi_1(X)\} + \pi_1(X)\mu_1(X) - e_0\{1 - \pi_0(X)\} - \pi_0(X)\mu_0(X) \Big]. \tag{9}$$

For the atom-at-zero representation used in most of the manuscript, Equation (9) reduces to $\mathbb{P}_n\{\pi_1(X)\mu_1(X) - \pi_0(X)\mu_0(X)\}$. These standardized quantities compare the two treatment arms at the same empirical covariate distribution, rather than comparing arm-specific fitted means over potentially different covariate distributions.

Proposition 1. *Assume that the parametric model for $Y_i \mid A_i = 1, R_i, X_i$ is correctly specified, that the regression models in Equations (3) and (4) are finite-dimensional and correctly specified, that the true parameter lies in the interior of the parameter space, that the model is identifiable with nonsingular Fisher information, and that $0 < \Pr(A_i = 1 \mid R_i = r, X_i) < 1$ almost surely for $r = 0, 1$. Let $n_r = \sum_{i=1}^{n} \mathbb{1}(R_i = r)$ and assume $n_0, n_1 \to \infty$ with $n_1/n_0 \to c \in (0, \infty)$. Then, under the null hypothesis $\mu_\delta = \beta_\delta = 0$, the profile likelihood-ratio statistic $W_n^P(0, 0)$ in Equation (5) converges in distribution to χ_2^2.*

Proof. Under the stated regularity conditions, the joint likelihood is a regular finite-dimensional likelihood with two scalar restrictions under the null: one in the continuous component and one in the Bernoulli component. Because the likelihood factorizes into $L_{Y,n}$ and $L_{A,n}$, the joint likelihood-ratio statistic is exactly the sum of the corresponding component likelihood-ratio statistics. Wilks' theorem (Wilks, 1938) therefore yields the χ_2^2 limit. □

The same fitted conditional laws determine both the component likelihood-ratio statistic and the standardized summaries in Equations (8)–(9). Thus covariate adjustment has two distinct roles: it defines the conditional restrictions tested by the likelihood ratio, and it yields marginal summaries by averaging the fitted arm-specific conditional quantities over a common empirical distribution of X_i.

2.3 Semi-parametric likelihood-ratio test

The parametric test depends on the chosen distributional model for the observed outcomes. To relax that assumption, we replace the Gaussian likelihood for the continuous non-atom outcome by an empirical likelihood for the conditional mean restrictions, while retaining the parametric Bernoulli likelihood for A_i. The observation component is therefore still governed by Equation (3), but no Normality, homoscedasticity, or parametric shape assumption is imposed on the distribution of $Y_i \mid A_i = 1, R_i, X_i$ beyond the finite-dimensional conditional mean model in Equation (4).

Let $\mathcal{J} = \{i : A_i = 1\}$ and $M_n = |\mathcal{J}|$. For $i \in \mathcal{J}$ define the observed-outcome design vector

$$Z_i = \{1, R_i, h_\mu(X_i)^\top\}^\top, \qquad \theta = (\mu_0, \mu_\delta, \eta_\mu^\top)^\top,$$

so that $\mu_{R_i}(X_i) = Z_i^\top \theta$. The semi-parametric continuous component is characterized by the moment restriction

$$\mathrm{E}\Big[Z_i\{Y_i - Z_i^\top \theta\} \mid A_i = 1\Big] = 0. \tag{10}$$

For a fixed candidate value m_δ of the treatment mean coefficient, write $\theta(m_\delta, \nu)$ for the vector obtained by setting the treatment component of θ equal to m_δ and collecting the remaining coefficients in the nuisance vector ν. Define

$$g_i\{m_\delta, \nu\} = Z_i \left[Y_i - Z_i^\top \theta(m_\delta, \nu)\right], \qquad i \in \mathcal{J}.$$

The profile empirical likelihood ratio for the continuous component is

$$R^{\mathrm{E}}_{Y,n}(m_\delta) = \sup_{\nu, \{p_i : i \in \mathcal{J}\}} \left\{ \prod_{i \in \mathcal{J}} M_n p_i \right\}, \tag{11}$$

where the supremum is over weights satisfying

$$p_i \geq 0, \qquad \sum_{i \in \mathcal{J}} p_i = 1, \qquad \sum_{i \in \mathcal{J}} p_i g_i\{m_\delta, \nu\} = 0.$$

The corresponding empirical-likelihood deviance is

$$\widetilde{W}^P_{Y,n}(m_\delta) = -2 \log R^{\mathrm{E}}_{Y,n}(m_\delta). \tag{12}$$

Equivalently, for each feasible (m_δ, ν) there is a Lagrange multiplier $\lambda(m_\delta, \nu)$ of the same dimension as Z_i such that

$$\sum_{i \in \mathcal{J}} \frac{g_i\{m_\delta, \nu\}}{1 + \lambda(m_\delta, \nu)^\top g_i\{m_\delta, \nu\}} = 0, \qquad 1 + \lambda(m_\delta, \nu)^\top g_i\{m_\delta, \nu\} > 0 \quad \text{for all } i \in \mathcal{J}.$$

In dual form,

$$\widetilde{W}^P_{Y,n}(m_\delta) = \inf_\nu 2 \sum_{i \in \mathcal{J}} \log \left[1 + \lambda(m_\delta, \nu)^\top g_i\{m_\delta, \nu\}\right], \tag{13}$$

with the value set to infinity when the empirical-likelihood constraints are infeasible. The fitted non-atom mean functions are standardized in the same way as in Equation (8):

$$\bar{\mu}^{\mathrm{SP}}_{r,n}(\theta) = \mathbb{P}_n\{(1, r, h_\mu(X)^\top)\theta\}, \qquad r \in \{0, 1\}.$$

Under the additive mean model, $\bar{\mu}^{\mathrm{SP}}_{1,n}(\theta) - \bar{\mu}^{\mathrm{SP}}_{0,n}(\theta) = \mu_\delta$. The semi-parametric standardized combined-outcome contrast is obtained by inserting this empirical-likelihood mean model and the fitted logistic observation probabilities into Equation (9). Thus the continuous distribution itself is left unspecified, but the mean component entering the standardized contrast is defined by the estimating equations in Equation (10).

When X_i is absent, the empirical-likelihood regression construction reduces to the familiar two-sample empirical likelihood for the observed outcome means. Let

$$\mathcal{I}_0 = \{i = 1, \ldots, n : R_i = 0, A_i = 1\}, \qquad \mathcal{I}_1 = \{i = 1, \ldots, n : R_i = 1, A_i = 1\}.$$

Lemma 4 in Appendix C shows that, for fixed (μ_0, μ_δ), the empirical-likelihood deviance for the observed outcomes is

$$\widetilde{W}_{Y,n}(\mu_0, \mu_\delta) = 2\sum_{i=1}^{n}\left[(1 - R_i)\log\left(1 + \lambda_0(\widetilde{Y}_i^{\,0} - A_i\mu_0)\right) + R_i \log\left(1 + \lambda_1(\widetilde{Y}_i^{\,0} - A_i(\mu_0 + \mu_\delta))\right)\right], \tag{14}$$

where terms with $A_i = 0$ contribute zero because $\widetilde{Y}_i^{\,0} = 0$ in that case. Writing

$$n_0 = \sum_{i=1}^{n}(1 - R_i)A_i = |\mathcal{I}_0|, \qquad n_1 = \sum_{i=1}^{n} R_i A_i = |\mathcal{I}_1|,$$

the dual parameters λ_0 and λ_1 solve

$$\frac{1}{n_0}\sum_{i=1}^{n}(1 - R_i)\frac{\widetilde{Y}_i^{\,0} - A_i\mu_0}{1 + \lambda_0(\widetilde{Y}_i^{\,0} - A_i\mu_0)} = 0,$$
$$\frac{1}{n_1}\sum_{i=1}^{n} R_i\frac{\widetilde{Y}_i^{\,0} - A_i(\mu_0 + \mu_\delta)}{1 + \lambda_1(\widetilde{Y}_i^{\,0} - A_i(\mu_0 + \mu_\delta))} = 0.$$

The profile version is obtained by minimizing over the nuisance mean:

$$\widetilde{W}^P_{Y,n}(\mu_\delta) = \inf_{\mu_0 \in \mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, \mu_\delta).$$

The admissible values of μ_0 are those for which the corresponding constrained means lie in the convex hull of the observed outcomes in each group. For interior feasible means, the dual parameters λ_0 and λ_1 are determined by one-dimensional root equations with unique roots on their admissible intervals, and the profiled nuisance mean is then minimized over its feasible set. If a candidate constraint falls outside the convex-hull region, the corresponding empirical-likelihood statistic is treated as infinite.

The semi-parametric test statistic is then

$$\widetilde{W}^P_n(\mu_\delta, \beta_\delta) = \widetilde{W}^P_{Y,n}(\mu_\delta) + W^P_{A,n}(\beta_\delta). \tag{15}$$

The first term is the profile empirical-likelihood deviance in Equation (12), or equivalently Equation (14) in the no-covariate special case; the second term is the Bernoulli profile likelihood-ratio statistic from Equation (5).

Proposition 2 (Covariate-adjusted semi-parametric likelihood-ratio test)**.** *Consider the null hypothesis*

$$H_0: \ \mu_\delta = 0, \quad \beta_\delta = 0,$$

equivalently, under the additive adjustment models, no covariate-standardized difference in the non-atom mean and no covariate-specific log odds-ratio for observation. Assume that (Y_i, A_i, R_i, X_i) *are independent observations from a common law; that for some* $\varepsilon > 0$, $\varepsilon \le \Pr(R_i = 1 \mid X_i) \le 1 - \varepsilon$ *and* $\varepsilon \le \Pr(A_i = 1 \mid R_i = r, X_i) \le 1 - \varepsilon$ *almost surely for* $r = 0, 1$*; that the logistic model in Equation (3) and the conditional mean model in Equation (4) are correctly specified; that the true finite-dimensional parameters lie in the interior of their parameter spaces; that the Fisher information for the Bernoulli component is nonsingular; and that, for the empirical-likelihood component,* $\mathrm{E}\,\|Z_i\{Y_i - Z_i^\top \theta_0\}\|^{2+\eta} < \infty$ *for some* $\eta > 0$*, the covariance matrix of* $Z_i\{Y_i - Z_i^\top \theta_0\}$ *conditional on* $A_i = 1$ *is nonsingular, and the derivative matrix of the moment condition in Equation (10) has full column rank. Then, under* H_0*,*

$$\widetilde{W}_n^P(0,0) = \widetilde{W}_{Y,n}^P(0) + W_{A,n}^P(0) \ \xrightarrow{d} \ \chi_2^2.$$

The two degrees of freedom correspond to the two scalar restrictions tested jointly: $\mu_\delta = 0$ *in the continuous mean component and* $\beta_\delta = 0$ *in the observation component.*

Proof. The Bernoulli component is a regular finite-dimensional likelihood with one scalar restriction, so Wilks' theorem gives $W_{A,n}^P(0) \xrightarrow{d} \chi_1^2$. Conditional on the non-atom sample, the continuous component is an empirical likelihood for estimating equations with a single profiled scalar restriction. The empirical-likelihood Wilks theorem for estimating equations therefore gives $\widetilde{W}_{Y,n}^P(0) \xrightarrow{d} \chi_1^2$ under the stated moment, rank, and interiority conditions (Owen, 1991; Qin and Lawless, 1994). The Bernoulli score is a function of (A_i, R_i, X_i), whereas the empirical-likelihood estimating function has conditional mean zero given $(A_i = 1, R_i, X_i)$; hence the two components are asymptotically block-orthogonal. The sum in Equation (15) therefore has the χ_2^2 null limit. □

The adjusted parametric and semi-parametric analyses consequently target the same two-component treatment structure. Both use the logistic model for the probability of escaping the atom and both summarize treatment effects at the empirical covariate distribution through Equations (8)–(9). They differ in the continuous non-atom component: the parametric analysis uses a fully specified Gaussian conditional distribution with common residual variance, whereas the semi-parametric analysis uses only the regression moment restrictions in Equation (10) and leaves the remaining shape of the continuous distribution unspecified. The semi-parametric likelihood-ratio test is therefore less sensitive to non-Normality of the continuous outcomes, at the cost of relying more directly on large-sample empirical-likelihood approximations and on feasibility of the moment constraints.

2.4 Inference and interpretation

Confidence regions follow by inverting the corresponding likelihood-ratio statistic. For the parametric procedure, an asymptotic $(1-\alpha)100\%$ simultaneous confidence region for $(\mu_\delta, \beta_\delta)$ is

$$\mathrm{CI}_{1-\alpha}^{(\mu_\delta,\beta_\delta)} = \left\{(m,b) \in \mathbb{R}^2 : W_n^P(m,b) \le \chi_{2,1-\alpha}^2\right\}.$$

For the semi-parametric procedure, one replaces W_n^P by $\widetilde{W}_n^P$. Correspondingly, the marginal intervals are

$$\mathrm{CI}_{1-\alpha}^{\mu_\delta} = \left\{ m \in \mathbb{R} : W_{Y,n}^P(m) \le \chi^2_{1,1-\alpha} \right\} \quad \text{or} \quad \left\{ m \in \mathbb{R} : \widetilde{W}_{Y,n}^P(m) \le \chi^2_{1,1-\alpha} \right\},$$
$$\mathrm{CI}_{1-\alpha}^{\beta_\delta} = \left\{ b \in \mathbb{R} : W_{A,n}^P(b) \le \chi^2_{1,1-\alpha} \right\}.$$

Here $\chi^2_{q,1-\alpha}$ denotes the $(1-\alpha)$ quantile of a chi-square distribution with q degrees of freedom. We recommend reporting the simultaneous confidence region because it displays the joint uncertainty about the effect on the continuous and binary components.

Many applied reports also require an interval on the numerical combined-outcome scale. Such an interval concerns a derived scalar functional, denoted here by Δ or $\Delta^{(e_0)}$, rather than the two-component parameter itself. It therefore answers a different inferential question from the region for $(\mu_\delta, \beta_\delta)$: it summarizes the consequence of the atom and continuous components after the atom has been assigned the numerical value e_0 through the representation map c_{e_0}. This scalar summary is useful for reporting effects on the original numerical scale, but it is secondary to the two-component analysis because it necessarily collapses changes in the observation probability and changes in the non-atom mean into a single number.

To define the likelihood-based intervals uniformly, let

$$\mathcal{W}_n(m,b) = \begin{cases} W_n^P(m,b), & \text{for the parametric likelihood,} \\ \widetilde{W}_n^P(m,b), & \text{for the semi-parametric likelihood.} \end{cases} \tag{16}$$

For fixed component contrasts (m,b), let $\widehat{\pi}_r(x;b)$ denote the fitted observation probability in arm r obtained after profiling the Bernoulli nuisance parameters under the restriction $\beta_\delta = b$. Similarly, let $\widehat{\mu}_r(x;m)$ denote the fitted conditional non-atom mean in arm r obtained after profiling the continuous-component nuisance parameters under the restriction $\mu_\delta = m$. In the parametric analysis this profiling is with respect to the Gaussian likelihood; in the semi-parametric analysis it is with respect to the empirical-likelihood estimating-equation constraints. The profiled combined-outcome contrast induced by (m,b) and the atom representation c_{e_0} is

$$\begin{aligned} g_n^{(e_0)}(m,b) = \mathbb{P}_n\Big[& e_0\{1-\widehat{\pi}_1(X;b)\} + \widehat{\pi}_1(X;b)\widehat{\mu}_1(X;m) \\ & - e_0\{1-\widehat{\pi}_0(X;b)\} - \widehat{\pi}_0(X;b)\widehat{\mu}_0(X;m)\Big]. \end{aligned} \tag{17}$$

When there are no adjustment covariates and $e_0 = 0$, this reduces to

$$g_n^{(0)}(m,b) = \widehat{\pi}_1(b)\widehat{\mu}_1(m) - \widehat{\pi}_0(b)\widehat{\mu}_0(m),$$

where $\widehat{\pi}_r(b)$ and $\widehat{\mu}_r(m)$ are the corresponding arm-specific profiled probabilities and means. The fitted likelihood-based estimate of the numerical combined-outcome contrast is $g_n^{(e_0)}(\widehat{\mu}_\delta, \widehat{\beta}_\delta)$, where $(\widehat{\mu}_\delta, \widehat{\beta}_\delta)$ denotes the unrestricted component estimate.

The first interval is the direct Welch interval for the atom-at-zero numerical outcome. Let $n_r = \sum_{i=1}^n \mathbb{1}(R_i = r)$ and define

$$\widehat{\Delta}_{\mathrm{W}} = \bar{\widetilde{Y}}_1^{\,0} - \bar{\widetilde{Y}}_0^{\,0}, \qquad \bar{\widetilde{Y}}_r^{\,0} = n_r^{-1} \sum_{i:R_i=r} \widetilde{Y}_i^{\,0},$$

with arm-specific sample variances

$$S_r^2 = (n_r - 1)^{-1} \sum_{i:R_i=r} \left(\widetilde{Y}_i^{\,0} - \bar{\widetilde{Y}}_r^{\,0} \right)^2, \qquad r \in \{0,1\}.$$

The Welch standard error and Satterthwaite degrees of freedom are

$$\widehat{\mathrm{se}}_{\mathrm{W}}^2 = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}, \qquad \widehat{\nu} = \frac{\left(S_1^2/n_1 + S_0^2/n_0\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_0^2/n_0)^2}{n_0-1}},$$

with the usual limiting interpretation if one of the variance terms is zero. The resulting interval is

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{W}} = \left[\widehat{\Delta}_{\mathrm{W}} - t_{\widehat{\nu},1-\alpha/2}\widehat{\mathrm{se}}_{\mathrm{W}}, \widehat{\Delta}_{\mathrm{W}} + t_{\widehat{\nu},1-\alpha/2}\widehat{\mathrm{se}}_{\mathrm{W}} \right], \tag{18}$$

where $t_{\widehat{\nu},1-\alpha/2}$ is the corresponding Student quantile. This interval treats $\widetilde{Y}^{\,0}$ as an ordinary real-valued outcome and therefore does not use the two-component likelihood structure.

The second interval is obtained by projecting the simultaneous likelihood-ratio confidence region for the component parameter onto the $\Delta^{(e_0)}$ scale. Define

$$\mathcal{C}_{2,n}(1-\alpha) = \left\{ (m,b) \in \mathbb{R}^2 : \mathcal{W}_n(m,b) \le \chi^2_{2,1-\alpha} \right\}. \tag{19}$$

The projected likelihood-ratio interval is the image of this region under the profiled map in Equation (17):

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{proj}} = \left[\inf_{(m,b)\in\mathcal{C}_{2,n}(1-\alpha)} g_n^{(e_0)}(m,b), \sup_{(m,b)\in\mathcal{C}_{2,n}(1-\alpha)} g_n^{(e_0)}(m,b) \right]. \tag{20}$$

Equivalently, its endpoints are the constrained optimization problems

$$L_{\mathrm{proj}} = \inf_{m,b} g_n^{(e_0)}(m,b) \quad \text{subject to} \quad \mathcal{W}_n(m,b) \le \chi^2_{2,1-\alpha}, \tag{21}$$

$$U_{\mathrm{proj}} = \sup_{m,b} g_n^{(e_0)}(m,b) \quad \text{subject to} \quad \mathcal{W}_n(m,b) \le \chi^2_{2,1-\alpha}. \tag{22}$$

Because this construction preserves the entire two-dimensional confidence set before mapping it to a scalar, it inherits simultaneous coverage for $(\mu_\delta, \beta_\delta)$ and is typically conservative for $\Delta^{(e_0)}$ alone.

The third interval profiles the likelihood ratio directly over the scalar constraint $\Delta^{(e_0)} = d$. For each $d \in \mathbb{R}$, define

$$T_n(d) = \inf_{m,b} \left\{ \mathcal{W}_n(m,b) : g_n^{(e_0)}(m,b) = d \right\}. \tag{23}$$

Thus evaluating $T_n(d)$ is itself a constrained optimization problem: minimize the joint likelihood-ratio surface over component contrasts whose profiled numerical combined-outcome contrast is exactly d. The profile likelihood-ratio interval is

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{prof}} = \left\{ d \in \mathbb{R} : T_n(d) \le \chi^2_{1,1-\alpha} \right\}, \tag{24}$$

with endpoints

$$L_{\mathrm{prof}} = \inf\{d : T_n(d) \le \chi^2_{1,1-\alpha}\}, \qquad U_{\mathrm{prof}} = \sup\{d : T_n(d) \le \chi^2_{1,1-\alpha}\}.$$

The one-degree-of-freedom calibration reflects that the null constraint is on the single scalar functional $g_n^{(e_0)}(m,b)$, while the remaining component direction is profiled out.

Proposition 3 (Likelihood-ratio intervals for the combined-outcome contrast)**.** *Let $\theta_\delta = (\mu_\delta, \beta_\delta)^\top$. Suppose that $\mathcal{W}_n$ satisfies the regularity conditions for the relevant parametric or semi-parametric likelihood-ratio theory in a neighborhood of the true value $\theta_{\delta 0}$. Assume further that $g_n^{(e_0)}$ converges uniformly in probability on this neighborhood to a continuously differentiable population map $g^{(e_0)}$, that $\nabla g^{(e_0)}(\theta_{\delta 0}) \neq 0$, and that $\Delta_0 = g^{(e_0)}(\theta_{\delta 0})$. Then*

$$T_n(\Delta_0) \xrightarrow{d} \chi_1^2,$$

and the profile interval in Equation (24) has asymptotic coverage $1-\alpha$ for Δ_0. Moreover, if $\mathcal{C}_{2,n}(1-\alpha)$ is an asymptotic $(1-\alpha)100\%$ confidence region for $\theta_{\delta 0}$, then the projected interval in Equation (20) has at least asymptotic coverage $1-\alpha$ for Δ_0.

Proof. The likelihood-ratio surface is locally quadratic under the regularity conditions for the corresponding parametric likelihood or empirical-likelihood estimating-equation model. Since $g^{(e_0)}$ is a smooth scalar map with nonzero gradient at $\theta_{\delta 0}$, the null hypothesis $g^{(e_0)}(\theta_\delta) = \Delta_0$ imposes one regular scalar restriction on the two-component parameter. Wilks' theorem and the empirical-likelihood analogue therefore give the χ_1^2 limit for the profiled statistic. For the projected interval, the event $\theta_{\delta 0} \in \mathcal{C}_{2,n}(1-\alpha)$ implies, by construction and continuity of $g_n^{(e_0)}$, that $g_n^{(e_0)}(\theta_{\delta 0})$ lies in the image interval up to an asymptotically negligible error. The stated coverage follows from the coverage of the two-dimensional confidence region. □

The three intervals therefore describe related but distinct targets. The Welch interval in Equation (18) is a direct one-dimensional interval for the mean difference of the coded numerical outcome $\widetilde{Y}^0$; it is simple and relatively model-light, but it does not distinguish whether the contrast arises from the atom probability, the continuous non-atom component, or both. The projected interval in Equation (20) is a likelihood-based summary derived from the simultaneous two-component confidence region; it is appropriate when $\Delta^{(e_0)}$ is reported as a secondary scalar summary while maintaining coherence with the joint analysis, and it is usually wider because it retains simultaneous uncertainty in both μ_δ and β_δ. The profile interval in Equation (24) is instead a direct likelihood-ratio interval for the scalar functional $\Delta^{(e_0)}$; it profiles over nuisance component directions and is therefore more focused on the numerical combined-outcome contrast. Thus the profile and projected intervals remain tied to the two-part likelihood decomposition, whereas the Welch interval is an analysis of the numerical representation alone.

2.5 Bayesian two-group comparison

The preceding likelihood-based development admits a natural Bayesian counterpart. We again suppress X_i and work in the no-covariate setting, because this is the clearest setting in which to formulate the two-part observed-data law itself. Let $\widetilde{Y}_i^{(e_0)} = c_{e_0}(\widetilde{Y}_i)$ denote the numerical representation of the combined outcome for a fixed atom value $e_0 \in \mathbb{R}$. In the two-group comparison without covariates, the primary objects are the arm-specific observed-data laws. It is therefore more natural to fit treatment-stratified models than to impose a common mixture across treatment arms. A common mixture with only arm-specific shifts or scales would encode a priori similarity in distributional shape across arms, whereas in the present setting treatment may affect both the atom probability and the shape of the continuous non-atom distribution. Separate arm-specific mixtures align directly with the scientific question of whether the full observed-data law differs between treatment groups.

Let $\mathcal{C} \subseteq \mathbb{R}$ denote the support of the continuous non-atom outcomes on the chosen numerical scale. In the two implementations considered below, $\mathcal{C} = \mathbb{R}$ or $\mathcal{C} = (0, \infty)$. Let $\lambda_{\mathcal{C}}$ denote Lebesgue measure on $\mathcal{C}$, and define the dominating measure

$$\nu_{e_0} = \delta_{e_0} + \lambda_{\mathcal{C}}.$$

For each treatment arm $r \in \{0, 1\}$, we write the law of the numerical combined outcome as

$$\mathcal{L}(\widetilde{Y}_i^{(e_0)} \mid R_i = r) = \rho_r \delta_{e_0} + \pi_r F_r^c, \qquad \pi_r = 1 - \rho_r, \qquad F_r^c \ll \lambda_{\mathcal{C}}, \tag{25}$$

where $\rho_r = \Pr(\widetilde{Y}_i^{(e_0)} = e_0 \mid R_i = r)$ is the arm-specific atom probability and F_r^c is an absolutely continuous probability measure on $\mathcal{C}$ describing the non-atom outcomes. Equivalently, with f_r^c denoting the density of F_r^c with respect to $\lambda_{\mathcal{C}}$, the density of $\widetilde{Y}_i^{(e_0)} \mid R_i = r$ with respect to ν_{e_0} is

$$q_r(y) = \rho_r \mathbb{1}\{y = e_0\} + \pi_r f_r^c(y) \mathbb{1}\{y \in \mathcal{C}\}.$$

One could instead start directly from a single density with respect to the extended measure ν_{e_0} on $\mathcal{C} \cup \{e_0\}$. For the present problem, however, the explicit singular-plus-absolutely-continuous decomposition in Equation (25) is clearer, because it preserves the atom probability and the non-atom distribution as separate scientific objects and mirrors the observed-data factorization used in the likelihood-based analysis.

The main modeling task is therefore to specify a flexible prior for F_r^c. This is precisely the setting in which a Dirichlet process mixture model is attractive. The conditional non-atom distribution may be skewed, heavy-tailed, or multimodal, and a single parametric family can be too restrictive even when the scientific summaries of interest remain simple. By modeling F_r^c through a nonparametric mixture, one can accommodate complex distributional shape while retaining interpretable treatment-arm summaries such as the atom probability, the mean among non-atom outcomes, and contrasts derived from them.

Let $(\Theta, \mathcal{T})$ denote the parameter space for a support-matched kernel $K(\cdot \mid \theta)$ on $\mathcal{C}$. An arm-specific mixing distribution G_r is said to follow a Dirichlet process with concentration parameter $\alpha_r > 0$ and base measure G_0, written $G_r \sim \mathrm{DP}(\alpha_r, G_0)$, if for every measurable partition $(B_1, \ldots, B_m)$ of Θ,

$$(G_r(B_1), \ldots, G_r(B_m)) \sim \mathrm{Dirichlet}(\alpha_r G_0(B_1), \ldots, \alpha_r G_0(B_m))$$

(Ferguson, 1973). When G_r is used as a mixing distribution,

$$F_r^c(B) = \int_\Theta K(B \mid \theta)\, G_r(d\theta), \qquad B \subseteq \mathcal{C},$$

which yields the familiar Dirichlet process mixture model (Antoniak, 1974). By Sethuraman's stick-breaking representation (Sethuraman, 1994),

$$G_r = \sum_{h=1}^{\infty} w_{rh} \delta_{\theta_{rh}}, \qquad w_{r1} = v_{r1}, \qquad w_{rh} = v_{rh} \prod_{\ell=1}^{h-1} (1 - v_{r\ell}), \quad h \geq 2,$$

with

$$v_{rh} \mid \alpha_r \sim \mathrm{Beta}(1, \alpha_r), \qquad \theta_{rh} \overset{\mathrm{iid}}{\sim} G_0.$$

For computation we use the standard finite truncated approximation (Ishwaran and James, 2001), retaining H mixture components and assigning the remaining stick mass to the last retained component. This is the approximation used in the practical implementation.

Writing $\widetilde{Y}_i^{(e_0)} = e_0$ when $A_i = 0$ and $\widetilde{Y}_i^{(e_0)} = Y_i$ when $A_i = 1$, the implemented no-covariate model may be expressed hierarchically as

$$
\begin{aligned}
A_i \mid R_i = r &\sim \text{Bernoulli}(\pi_r), \qquad r \in \{0,1\}, \\
\pi_r &= 1 - \rho_r, \qquad \rho_r \sim \text{Beta}(a_\rho, b_\rho), \\
Y_i \mid (A_i = 1, R_i = r) &\sim F_r^c, \\
F_r^c(dy) &= \sum_{h=1}^{H} w_{rh} K(dy \mid \theta_{rh}), \\
v_{rh} \mid \alpha_r &\sim \text{Beta}(1, \alpha_r), \qquad h = 1, \ldots, H-1, \\
\alpha_r &\sim \text{Gamma}(a_\alpha, b_\alpha), \\
w_{r1} &= v_{r1}, \\
w_{rh} &= v_{rh} \prod_{\ell=1}^{h-1} (1 - v_{r\ell}), \qquad h = 2, \ldots, H-1, \\
w_{rH} &= \prod_{\ell=1}^{H-1} (1 - v_{r\ell}), \\
K(dy \mid \theta_{rh}) &= \phi(y; m_{rh}, s_{rh}^2)\, dy, \qquad \mathcal{C} = \mathbb{R}, \\
m_{rh} &\sim N(m_0, s_0^2), \qquad s_{rh} \sim \text{Lognormal}(m_s, s_s^2), \\
K(dy \mid \theta_{rh}) &= g(y; a_{rh}, a_{rh}/m_{rh})\ dy, \qquad \mathcal{C} = (0, \infty), \\
m_{rh} &\sim \text{Lognormal}(m_m, s_m^2), \qquad a_{rh} \sim \text{Lognormal}(m_a, s_a^2).
\end{aligned}
\tag{26}
$$

where $\phi(\cdot; m, s^2)$ denotes the Normal density with mean m and variance s^2, and $g(\cdot; a, b)$ denotes the Gamma density with shape a and rate b. Thus the full-real-line version uses Normal kernels, whereas the positive-support version uses Gamma kernels parameterized so that m_{rh} is the component mean. The latter formulation requires all non-atom outcomes to be strictly positive. In either case, the mixture is arm-specific, the atom probability is explicit, and the scientific summaries remain directly interpretable despite the flexibility of the nonparametric continuous component.

Posterior inference. Let $\mathcal{D}_n = \{(\widetilde{Y}_i^{(e_0)}, A_i, R_i) : i = 1, \ldots, n\}$ denote the observed data, and let ϑ collect all unknown quantities in Equation (26). For arm r, the posterior predictive distribution of a future numerical combined outcome $\widetilde{Y}_{r,\text{new}}^{(e_0)}$ is

$$
\begin{aligned}
\Pr\Big(\widetilde{Y}_{r,\text{new}}^{(e_0)} \in B \mid \mathcal{D}_n\Big) = \int \Bigg\{ & \rho_r \delta_{e_0}(B) \\
& + \pi_r \int_{B \cap \mathcal{C}} f_r^c(y \mid \vartheta)\, \lambda_{\mathcal{C}}(dy) \Bigg\} p(\vartheta \mid \mathcal{D}_n)\, d\vartheta.
\end{aligned}
$$

where

$$
f_r^c(y \mid \vartheta) = \sum_{h=1}^{H} w_{rh} k(y \mid \theta_{rh})
$$

is the arm-specific continuous mixture density. For posterior draw m, the corresponding mean among non-atom outcomes is

$$
\mu_r^{c,(m)} = \int_{\mathcal{C}} y f_r^c(y \mid \vartheta^{(m)})\, \lambda_{\mathcal{C}}(dy),
$$

which under both implemented kernels reduces to the weighted component-mean average $\mu_r^{c,(m)} = \sum_{h=1}^{H} w_{rh}^{(m)} m_{rh}^{(m)}$. The package reports the arm-specific summaries ρ_r, π_r, and μ_r^c together with the treatment contrasts

$$\Delta_{\text{atom}} = \rho_1 - \rho_0, \qquad \mu_\delta = \mu_1^c - \mu_0^c, \qquad \alpha_\delta = \frac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)},$$
$$\mu_r^{(e_0)} = e_0\rho_r + \pi_r\mu_r^c, \qquad \Delta^{(e_0)} = \mu_1^{(e_0)} - \mu_0^{(e_0)}. \tag{27}$$

For the atom-at-zero representation used elsewhere in the manuscript, $\Delta = \Delta^{(0)} = \pi_1\mu_1^c - \pi_0\mu_0^c$. Thus the Bayesian analysis reports, on a common posterior scale, the atom probability in each arm, the mean among non-atom outcomes in each arm, the atom-probability contrast, the contrast in non-atom means, the odds-ratio summary α_δ for escaping the atom, and the induced contrast on the numerical combined-outcome scale.

All of these functionals are computed draw-by-draw from the posterior sample and are then summarized by posterior medians or means, equal-tail credible intervals, and directional posterior probabilities. In the present implementation the default point summaries are posterior medians, with equal-tail credible intervals. Because several scientifically relevant quantities are nonlinear functions of (ρ_r, π_r, μ_r^c), it is important to compute them at each posterior draw rather than by plugging posterior means into the formulas in Equation (27). Directional summaries such as $\Pr(\Delta_{\text{atom}} < 0 \mid \mathcal{D}_n)$, $\Pr(\mu_\delta > 0 \mid \mathcal{D}_n)$, $\Pr(\alpha_\delta > 1 \mid \mathcal{D}_n)$, or $\Pr(\Delta^{(e_0)} > 0 \mid \mathcal{D}_n)$ are then immediate from the posterior draws, with the relevant direction chosen according to whether larger or smaller outcomes are clinically preferred.

Posterior predictive checks and model assessment. Model adequacy is assessed through posterior predictive checks in the sense of Gelman et al. (1996). These are Bayesian diagnostics based on the posterior predictive distribution, not frequentist goodness-of-fit tests. For each posterior draw $\vartheta^{(m)}$, one generates a replicated data set by first drawing

$$A_i^{\text{rep},(m)} \mid R_i = r, \vartheta^{(m)} \sim \text{Bernoulli}\left(\pi_r^{(m)}\right),$$

and then, for those replicated subjects with $A_i^{\text{rep},(m)} = 1$,

$$Y_i^{\text{rep},(m)} \mid \left(A_i^{\text{rep},(m)} = 1, R_i = r, \vartheta^{(m)}\right) \sim F_r^{c,(m)}.$$

The replicated combined outcome is finally reconstructed as $\widetilde{Y}_i^{\text{rep},(e_0,m)} = e_0$ when $A_i^{\text{rep},(m)} = 0$ and $\widetilde{Y}_i^{\text{rep},(e_0,m)} = Y_i^{\text{rep},(m)}$ otherwise. The observed data and the replicated data are then compared through discrepancies chosen to target the two components of the model.

For the atom part, the relevant question is whether the fitted Bernoulli model reproduces the observed armwise frequency of the atom. A natural discrepancy is therefore the maximum armwise absolute deviation between the observed atom rate and the fitted atom probability,

$$T_A(\widetilde{y}, \vartheta) = \max_{r \in \{0,1\}} \left| \frac{1}{n_r} \sum_{i: R_i = r} \mathbb{1}\{\widetilde{y}_i = e_0\} - \rho_r \right|.$$

For the continuous non-atom part, the relevant question is whether the fitted mixture reproduces the shape of the conditional distribution among subjects with observed outcomes. The implementation

therefore uses the maximum armwise Kolmogorov–Smirnov discrepancy between the empirical non-atom distribution and the fitted continuous mixture CDF,

$$T_C(y, \vartheta) = \max_{r \in \{0,1\}} \sup_{t \in \mathcal{C}} \left| \widehat{F}^c_{r,\mathrm{obs}}(t) - F^c_r(t \mid \vartheta) \right|.$$

For positive-support models the same comparison can be displayed on the log scale for graphical stability, but the target remains the arm-specific conditional non-atom law. These checks are supplemented naturally by graphical posterior predictive displays, such as grouped bar plots for the atom indicators and density or CDF overlays for the non-atom outcomes.

More generally, one may also examine scalar posterior predictive discrepancies based on arm-specific means, tail probabilities, quantiles, or other summaries chosen for the application. For any discrepancy $T(\cdot, \vartheta)$, the posterior predictive p-value is

$$p_{\mathrm{ppc}} = \Pr\left\{ T(\widetilde{Y}^{\mathrm{rep}}, \vartheta) \geq T(\widetilde{Y}, \vartheta) \,\middle|\, \mathcal{D}_n \right\},$$

estimated by the corresponding empirical proportion over posterior draws and replicated data sets. Such quantities are useful diagnostics for systematic lack of fit in either the atom model or the continuous conditional distribution, but they should be interpreted as Bayesian model-checking summaries rather than as classical significance tests.

Choosing the truncation level. Because the infinite stick-breaking mixture is implemented through a finite truncation, the number of retained mixture components H should be viewed as an approximation-control parameter rather than as an estimand for a "true" finite number of mixture components. The inferential target is the posterior of the scientific functionals in Equation (27), not the posterior of an exact component count. This distinction is important in flexible mixture models, because nearby kernels may split or merge without materially changing either the fitted non-atom distribution or the posterior of the scientific summaries.

For arm $r \in \{0, 1\}$, the truncated stick-breaking construction implies

$$v_{rh} \mid \alpha_r \sim \mathrm{Beta}(1, \alpha_r), \qquad h = 1, \ldots, H-1,$$

with weights

$$w_{r1} = v_{r1}, \qquad w_{rh} = v_{rh} \prod_{\ell=1}^{h-1} (1 - v_{r\ell}), \quad h = 2, \ldots, H-1,$$

and

$$w_{rH} = \prod_{\ell=1}^{H-1} (1 - v_{r\ell}).$$

Thus w_{rH} is the residual stick mass after the first $H-1$ breaks, and the implementation assigns this remaining mass to the final retained component. If one were to continue the stick-breaking construction by one additional break $V_{rH} \mid \alpha_r \sim \mathrm{Beta}(1, \alpha_r)$, then the mass lying strictly beyond the retained truncation point would be $w_{rH}(1 - V_{rH})$. Since

$$\mathrm{E}(1 - V_{rH} \mid \alpha_r) = \frac{\alpha_r}{1 + \alpha_r},$$

a natural truncation diagnostic is

$$T_{rH} = w_{rH} \frac{\alpha_r}{1 + \alpha_r}, \tag{28}$$

which is the posterior expected omitted stick mass beyond H, conditional on the currently retained residual stick. Small values of T_{rH} indicate that little posterior mass remains beyond the truncation boundary. This is more informative than monitoring w_{rH} alone, because the last retained component may itself be genuinely occupied by data.

We therefore choose H by monitoring the posterior of T_{rH} rather than by treating the number of occupied components as a scientific target. Let

$$n_r^{\text{obs}} = \sum_{i=1}^{n} \mathbb{1}(A_i = 1, R_i = r)$$

denote the number of observed non-atom outcomes in arm r, and define the arm-specific tolerance

$$\eta_r = \min\left\{0.02, \frac{1}{n_r^{\text{obs}}}\right\}.$$

Operationally, we regard a truncation level as adequate when, for both treatment arms,

$$Q_{0.95}(T_{rH} \mid \mathcal{D}_n) < \eta_r, \tag{29}$$

and the Markov chain Monte Carlo diagnostics are satisfactory, with no divergent transitions, $\widehat{R} \leq 1.01$, and adequate bulk and tail effective sample sizes. If Equation (29) fails in either arm, then H is increased and the model is refit; the final analysis uses the smallest H satisfying the criterion.

In the present implementation the ladder $H = 10 \to 20 \to 40$ is especially natural under the default prior $\alpha_r \sim \text{Gamma}(2, 1)$. Conditional on α_r,

$$\text{E}(w_{rH} \mid \alpha_r) = \left(\frac{\alpha_r}{1 + \alpha_r}\right)^{H-1},$$

so that

$$\text{E}(T_{rH}) = \text{E}\left[\left(\frac{\alpha_r}{1 + \alpha_r}\right)^{H}\right].$$

Under Gamma$(2, 1)$, this prior mean omitted mass is approximately 0.033 for $H = 10$, 0.0039 for $H = 20$, and 1.6×10^{-4} for $H = 40$. Thus $H = 10$ already provides a moderate truncation, $H = 20$ drives the prior expected omitted mass below one-half of one percent, and $H = 40$ makes prior truncation error negligible for practical purposes. Posterior predictive checks remain essential for overall model assessment, but they address model adequacy rather than truncation adequacy and therefore complement, rather than replace, the diagnostic in Equation (29).

Covariate extensions. The no-covariate treatment-stratified model is the primary development in this paper, but the same two-part architecture extends naturally when baseline covariates are present. A direct extension retains treatment stratification and replaces the arm-specific constants π_r and F_r^c by covariate-dependent quantities $\pi_r(x)$ and $F_r^c(\cdot \mid x)$. For example, one may write

$$\begin{aligned} A_i \mid R_i = r, X_i = x &\sim \text{Bernoulli}\big(\pi_r(x)\big), \\ Y_i \mid (A_i = 1, R_i = r, X_i = x) &\sim F_r^c(\cdot \mid x), \end{aligned}$$

and then let the component-specific kernel means vary with x within each treatment arm. A convenient dependent Dirichlet process construction is

$$Y_i \mid (A_i = 1, R_i = r, X_i = x, z_i = h) \sim K(\cdot \mid m_{rh}(x), \vartheta_{rh}),$$
$$m_{rh}(\cdot) \sim \mathcal{GP}(\mu_{0h}(\cdot), C_h(\cdot, \cdot)),$$

so that the kernel locations vary smoothly with the covariates while the random mixing distribution remains treatment-arm specific. This yields a predictor-dependent mixture model in the spirit of dependent Dirichlet process constructions (MacEachern, 1999; Gelfand et al., 2005); see also Rasmussen and Williams (2006) for Gaussian process priors. Such a formulation preserves the two-part interpretation while allowing treatment groups to differ not only in atom probability but also in how the entire non-atom distribution evolves over the covariates.

If a more modular regression component is preferred, the Gaussian-process mean functions can be replaced by Bayesian additive regression trees on the kernel means, thereby retaining a flexible regression surface while leaving the mixture structure unchanged (Chipman et al., 2010). This is attractive when nonlinear covariate effects and interactions are important but a fully predictor-dependent random measure is unnecessary.

When a covariate is purely categorical, especially when it indexes substantively related groups, a hierarchical construction is often more appropriate than a fully continuous predictor-dependent process. For levels $\ell = 1, \ldots, L$ of a factor covariate, one may introduce within each treatment arm

$$G_r^\star \sim \mathrm{DP}(\gamma_r, H_0), \qquad G_{r\ell} \mid G_r^\star \sim \mathrm{DP}(\alpha_r, G_r^\star),$$

so that

$$F_{r\ell}^c(B) = \int_\Theta K(B \mid \theta)\, G_{r\ell}(d\theta), \qquad B \subseteq \mathcal{C}.$$

This is a hierarchical Dirichlet process mixture model (Teh et al., 2006); ANOVA-type dependent random-measure formulations provide a closely related alternative for factor effects (De Iorio et al., 2004). If the factor levels correspond to different hospital sites, such a hierarchical construction is especially natural. Hospitals are expected to have related but not identical outcome distributions because of differences in case mix, clinical practice, or measurement processes. The site-specific random measures $G_{r\ell}$ therefore allow each hospital to retain its own atom probability model and its own non-atom distribution, while the shared parent measure $G_r^\star$ induces borrowing of strength by encouraging the hospitals within a treatment arm to reuse common mixture atoms. In this way, data-rich sites stabilize inference for data-sparse sites without forcing the site-specific distributions to be identical. These covariate-adjusted extensions are conceptually straightforward continuations of the no-covariate framework, but the basic two-group model already captures the main ideas needed for the present paper.

3 Simulation study

The simulation study had two purposes: to assess null calibration and to compare power with standard one-dimensional analyses when treatment acts through the atom, the positive part, or both. In every replicate, the two treatment groups had equal size, we worked on the atom-at-zero numerical representation, equivalently the coded variable $\widetilde{Y}_i^{\,0} = c_0(\widetilde{Y}_i)$, and each subject contributed the numerical combined outcome $\widetilde{Y}_i^{\,0} = A_i Y_i$, where $A_i \mid R_i = r$ was Bernoulli with scenario-specific probability π_r and $Y_i \mid (A_i = 1, R_i = r)$ followed the corresponding survivor distribution.

Table 1 lists the data-generating mechanisms. The scenarios were chosen to isolate the practically important comparisons: a low-atom normal reference case (Scenario 1), a pure atom-probability effect (Scenario 2), an antagonistic case in which the atom and continuous components move in opposite directions while the mean of the coded combined outcome stays fixed (Scenario 3), a concordant joint improvement case (Scenario 4), a gamma shape-change case with survivor-mean shift (Scenario 5), and a rare-outlier contamination case that is deliberately favorable to rank-based methods (Scenario 6).

Table 1: Simulation scenarios for the TruncComp2 power study

Scenario	Configuration	$P(A=1 \mid R)$	$Y \mid A=1, R$
S1	Normal survivor shift, low atom	$\pi_0=\pi_1=0.85$	$Y \mid A=1, R=0 \sim N(3,1)$, $Y \mid A=1, R=1 \sim N(3+\delta_h,1)$, $\delta_h \in \{0, 0.20, 0.35, 0.50\}$
S2	Survival-only effect	$\pi_0=0.45$, $\pi_1 \in \{0.45, 0.50, 0.55, 0.60\}$	$Y \mid A=1, R=0 \sim Y \mid A=1, R=1 \sim N(3,1)$
S3	Antagonistic cancellation	$\pi_0=0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$	$Y \mid A=1, R=0 \sim N(3,1)$, $Y \mid A=1, R=1 \sim N(1.5/\pi_1,1)$
S4	Concordant joint improvement	$\pi_0=0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$	$Y \mid A=1, R=0 \sim N(3,1)$, $Y \mid A=1, R=1 \sim N(3+0.15h,1)$
S5	Gamma shape-change with mean shift	$\pi_0=\pi_1=0.65$	$Y \mid A=1, R=0 \sim \Gamma(9,3/9)$, $Y \mid A=1, R=1 \sim \Gamma(k_h, m_h/k_h)$, $(k_h, m_h) \in \{(9,3.0),(6,3.2),(4,3.4),(2.5,3.6)\}$
S6	Rare-outlier contamination	$\pi_0=\pi_1=0.80$	$Y \mid A=1, R=r$ is $0.95N(3+\delta_h r, 0.35^2)+0.05N(10+\delta_h r, 0.5^2)$, $\delta_h \in \{0, 0.10, 0.20, 0.35\}$

For each scenario we considered effect levels $h \in \{0,1,2,3\}$, with $h=0$ as the null configuration, and sample sizes from 50 to 500 observations per group in increments of 50. This yielded $6 \times 4 \times 10 = 240$ scenario-effect-sample-size cells, and each cell used 10,000 Monte Carlo repetitions. Power and empirical size were estimated as rejection proportions at the 5% level, so the worst-case Monte Carlo standard error of an estimated rejection rate was approximately 0.005. We compared the parametric and semi-parametric likelihood-ratio tests with Welch's two-sample t-test and the Wilcoxon rank-sum test applied directly to the combined outcome; Appendix A records the exact simulation protocol and implementation details needed to reproduce the study.

Figure 1 shows power as a function of sample size at the strongest non-null level across all six scenarios. Figure 2 complements that display by plotting power as a function of the effect level at $n=250$ observations per group across the same six scenarios. The supplementary material reports the full scenario-specific power tables for all 240 cells and the null-calibration displays for the $h=0$ settings.

The results support the main motivation for the proposed tests, but they also show that no single procedure is uniformly best. Scenario 3 is the clearest verification of the two-component idea. In that antagonistic setting, the coded combined-outcome mean is fixed at every effect level even though the atom probability and survivor mean both change. Consequently, at $n=250$ and the strongest effect level, the one-dimensional tests remain close to their null rejection rates, while both likelihood-ratio procedures reject essentially all replicates. This is the setting in which decomposing the atom and non-atom components is indispensable: the treatment effect is real in the observed-data law, but it has been deliberately hidden from a test of the numerical combined mean.

The other scenarios clarify when this advantage should and should not be expected. In the normal reference setting (Scenario 1), the parametric and semi-parametric likelihood-ratio tests track each other closely and have high power, as expected under a correctly specified continuous component. In the survival-only setting (Scenario 2), however, the likelihood-ratio procedures do not dominate the one-dimensional tests; at the strongest effect level and $n=250$, all four methods have power around 86–87%. The gain from modeling the Bernoulli component directly is offset by the two-degree-of-freedom nature of the joint test in this particular configuration. In the concordant setting (Scenario 4), where both components improve in the same direction, all methods perform well because the treatment effect is already visible on the coded combined-outcome scale.

The non-normal scenarios illustrate the limits of the procedures and the role of the survivor dis-

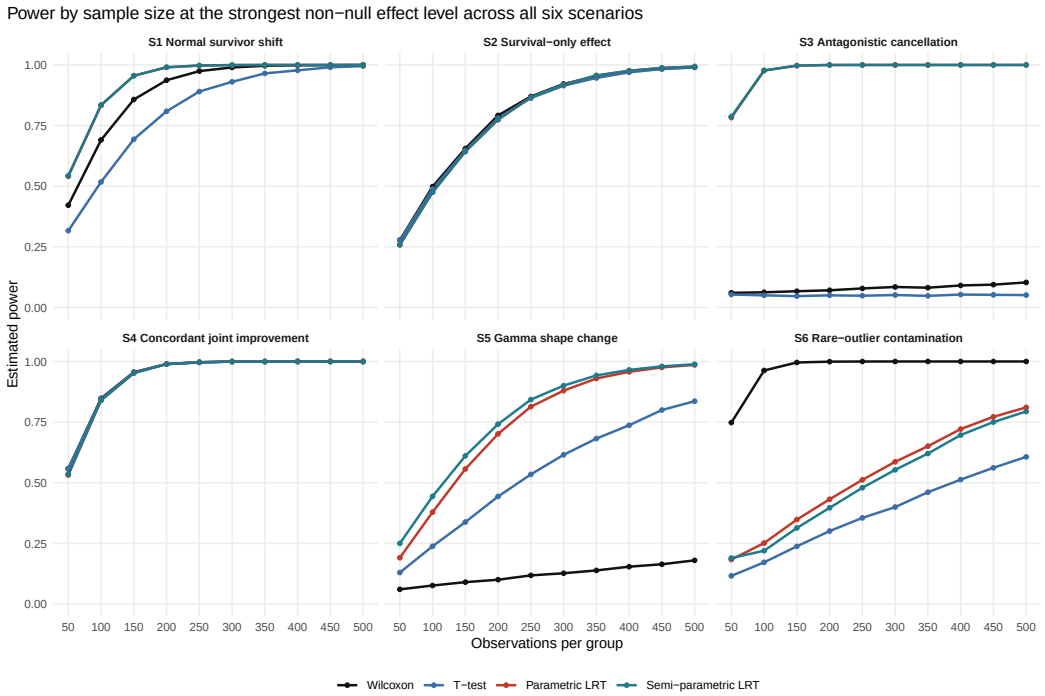

Figure 1: Estimated power as a function of sample size at the strongest non-null level for Scenarios 1 through 6, displayed in a 2×3 layout.

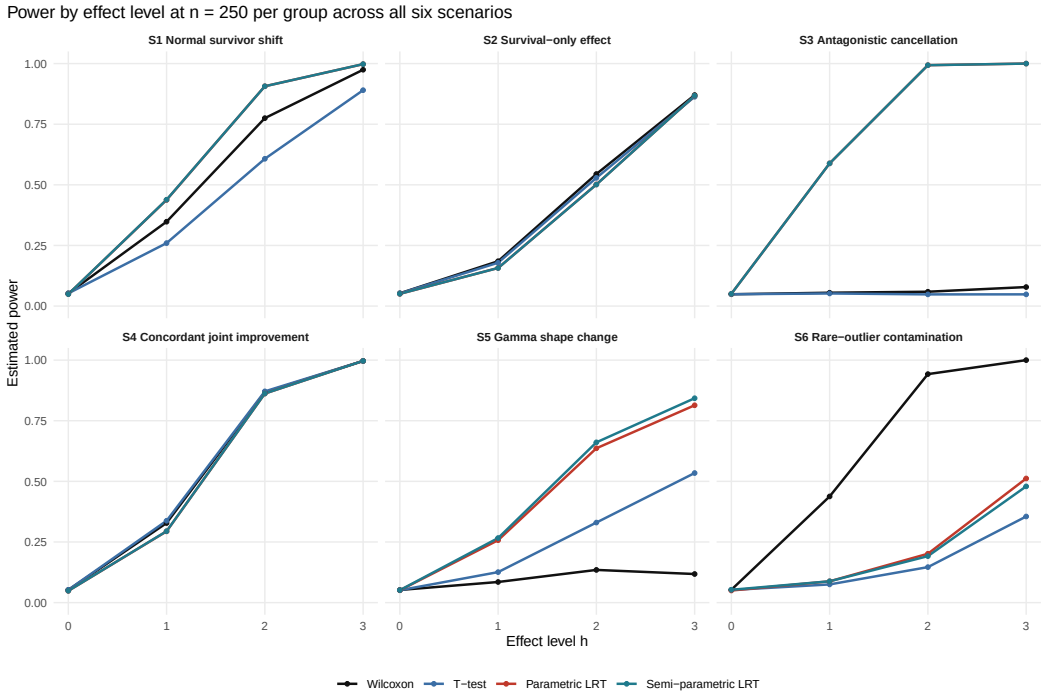

Figure 2: Estimated power as a function of effect level at $n = 250$ observations per group for Scenarios 1 through 6, displayed in a 2×3 layout.

tribution. In the gamma shape-change setting (Scenario 5), the semi-parametric likelihood-ratio test is modestly more powerful than the parametric version and both substantially outperform the one-dimensional tests, reflecting the benefit of a method that targets the survivor mean without imposing a normal positive-part distribution. In the rare-outlier contamination setting (Scenario 6), the rank-based Wilcoxon test is intentionally favored and has the highest power, whereas the proposed likelihood-ratio tests remain more sensitive than the t-test but are not competitive with Wilcoxon at the smaller sample sizes. This scenario is a useful reminder that the proposed methods are designed for component-specific mean and atom effects, not for every possible distributional alternative.

The null-calibration results are mostly reassuring but not perfectly uniform. The t-test, Wilcoxon test, and parametric likelihood-ratio test remain close to the nominal 5% level across the null configurations. The semi-parametric likelihood-ratio test is also close to nominal except in the smallest rare-outlier null cells, where it is anti-conservative at $n = 50$ and still mildly elevated at $n = 100$ before settling near 5% for larger samples. Thus the main conclusions are strongest for moderate sample sizes, while very small contaminated samples require caution for the semi-parametric procedure. The long supplementary tables document the full scenario-by-sample-size patterns underlying Figure 1.

4 Clinical application: IST-3 stroke trial

We apply the proposed methods to the third International Stroke Trial (IST-3), a randomized trial of intravenous alteplase (rt-PA) versus control in acute ischemic stroke (IST-3 Collaborative Group, 2012; Sandercock et al., 2016). Acute ischemic stroke is a natural setting for atom-plus-continuous endpoints: treatment may affect both mortality and functional health, and death before follow-up is not a missing quality-of-life value but a clinically meaningful terminal state. Among survivors, the 6-month EuroQol visual analogue score (EQ-VAS) is a patient-centered continuous measure on a 0 to 100 scale, with larger values indicating better self-rated health.

The IST-3 data were downloaded from Edinburgh DataShare. The fixed-width data file `ist3.dat` and accompanying SAS syntax identify allocated treatment, death by 6 months, and 6-month EQ-VAS. The syntax labels `itt_treat` as allocated treatment, `dead6mo` as death before 6 months, and `euroqol6` as the 6-month EuroQol health state. We verified the treatment coding against the text treatment field: `itt_treat = 0` corresponds to rt-PA and `itt_treat = 1` corresponds to placebo/control. We therefore set $R = 1$ for rt-PA and $R = 0$ for placebo/control. The DataShare record is publicly downloadable after the end of the embargo, but reuse is conditional on the IST-3 Data Use Agreement; the data citation is Sandercock et al. (2016), DOI 10.7488/ds/1350.

For the application endpoint, we code

$$Y = \begin{cases} -1, & \text{death by 6 months,} \\ \text{6-month EQ-VAS,} & \text{alive with observed EQ-VAS,} \end{cases} \qquad A = \mathbb{1}(Y \neq -1).$$

Thus the atom is -1, representing death by 6 months, and the continuous component is EQ-VAS among 6-month survivors. Alive participants with missing 6-month EQ-VAS were excluded from this analysis because the continuous component was not observed; deaths were retained through the atom. This left 2,733 participants for analysis, including 1,361 placebo/control participants

and 1,372 rt-PA participants. Table 2 reports the analysis sample, death counts, survivor EQ-VAS summaries, and the combined endpoint mean.

Table 2: IST-3 analysis sample and endpoint summaries.

Summary	Placebo/control	rt-PA
Randomized participants	1,520	1,515
Included in analysis	1,361	1,372
Alive but missing 6-month EQ-VAS	159	143
Death atom, n (%)	407 (29.9%)	408 (29.7%)
Alive with EQ-VAS, n (%)	954 (70.1%)	964 (70.3%)
Survivor EQ-VAS, mean (SD)	57.9 (23.8)	60.7 (23.3)
Survivor EQ-VAS, median (IQR)	60 (40, 75)	63 (49, 80)
Survivor EQ-VAS, range	0 to 100	0 to 100
Combined endpoint mean	40.3	42.3

Figure 3 displays the atom and survivor components together with the semiparametric joint confidence region. The death atom was similar in the two arms: 407 placebo/control participants (29.9%) and 408 rt-PA participants (29.7%) died before 6 months. Among survivors with EQ-VAS, however, the mean was 57.9 in the placebo/control arm and 60.7 in the rt-PA arm, giving $\hat{\mu}_\delta = 2.82$ points. The fitted odds ratio for being alive with observed EQ-VAS rather than at the death atom was $\hat{\alpha}_\delta = 1.008$, and the combined mean contrast under the $Y = -1$ death coding was $\hat{\Delta} = 2.08$.

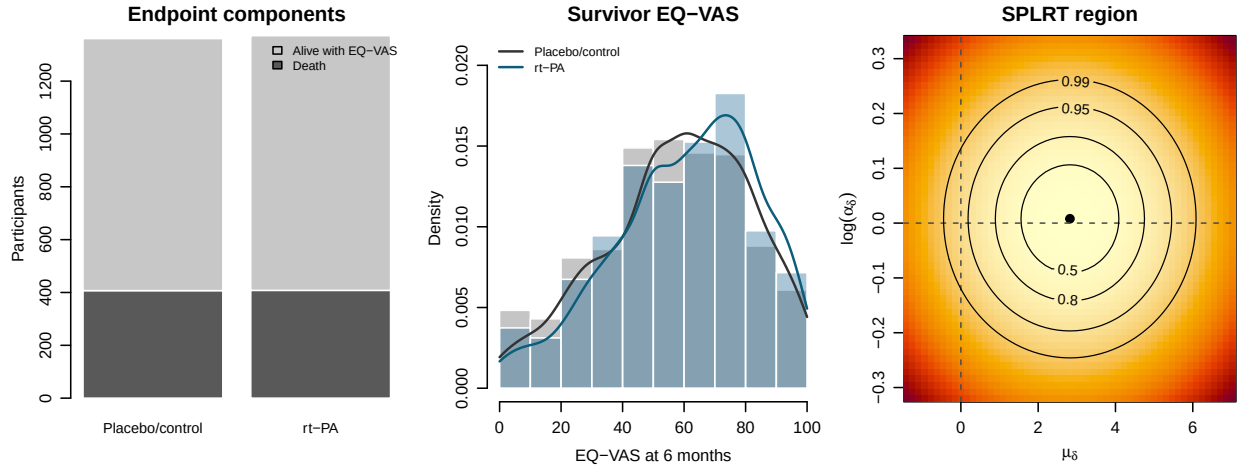

Figure 3: IST-3 endpoint components and semiparametric likelihood-ratio region. The left panel shows death and alive-with-EQ-VAS counts by treatment arm. The middle panel shows the survivor EQ-VAS distribution. The right panel shows the semiparametric likelihood-ratio surface for $(\mu_\delta, \log \alpha_\delta)$ with reference lines at no survivor mean difference and no atom-probability difference.

Table 3 compares standard one-dimensional analyses with the proposed joint tests. A Welch t-test applied directly to the atom-coded endpoint gave p-value 0.109, and the Wilcoxon rank-sum test gave p-value 0.094. The atom-coded t-test depends on the numerical choice -1 for death; moving the death atom would change the mean contrast and its standard error. The Wilcoxon test is less tied to that exact value, but it treats all deaths as tied observations at the worst rank and still collapses the endpoint to one dimension. A survivor-only analysis is more sensitive to the EQ-VAS difference, but it conditions on a post-randomization survival event and therefore answers

a different observed-data question.

Table 3: Standard and likelihood-ratio analyses of the IST-3 endpoint.

Analysis	Method	Estimate/target	Statistic	p-value
Combined endpoint	Welch t-test	$\Delta = 2.08$	-1.60	0.109
Combined endpoint	Wilcoxon rank-sum	Death tied as worst value	899,629.50	0.094
Survivors only	Welch t-test	$\mu_\delta = 2.82$	-2.62	0.009
Survivors only	Wilcoxon rank-sum	Conditional on survival	427,369.50	0.007
Death atom	Fisher exact test	Death proportions		0.933
Parametric model	Joint likelihood-ratio test	$\mu_\delta = 2.82$, $\alpha_\delta = 1.008$, $\Delta = 2.08$	6.88	0.032
Semiparametric model	Joint likelihood-ratio test	$\mu_\delta = 2.82$, $\alpha_\delta = 1.008$, $\Delta = 2.08$	6.87	0.032

The likelihood-ratio procedures test the joint null of no treatment effect on the death atom probability and no treatment effect on survivor EQ-VAS. The parametric LRT statistic was 6.88 with p-value 0.032, and the semiparametric LRT statistic was 6.87 with p-value 0.032. In this data set the two procedures agree closely, suggesting that the signal is mainly a location shift in the survivor EQ-VAS component rather than a feature driven by the parametric distributional assumption. Clinically, μ_δ is the rt-PA minus control mean EQ-VAS difference among 6-month survivors; α_δ is the rt-PA/control odds ratio for being alive with observed EQ-VAS rather than at the death atom; and Δ is the rt-PA minus control mean of the combined endpoint after assigning death the value -1. The joint rejection should therefore be interpreted as evidence that the observed two-component endpoint distribution differs between randomized arms, not as a survivor-average causal effect.

Table 4: Confidence intervals for the combined mean contrast Δ from the semiparametric IST-3 fit.

Interval construction	Lower	Upper	Status
Welch	-0.46	4.63	Estimated
Profile likelihood	-0.46	4.62	Estimated
Projected joint region	-1.09	5.25	Estimated

Table 4 illustrates three confidence intervals for Δ implemented in the software. The Welch interval is the direct interval for the atom-coded mean contrast. The profile interval constrains the full likelihood at a fixed Δ, while the projected interval maps the joint confidence region for the component parameters into the implied combined mean contrast. In IST-3, the profile interval is nearly identical to the Welch interval, whereas the projected interval is wider because it preserves joint uncertainty in both the atom and survivor components.

We also fit the experimental Bayesian two-part model in `TruncComp2`. The Bernoulli atom probabilities used independent Beta$(1, 1)$ priors, and the continuous survivor EQ-VAS distribution used a two-component normal mixture on the internally standardized real-line scale with weakly informative component and concentration priors. The real-line model was used because some survivors had EQ-VAS equal to 0, so a positive-support model would be inappropriate. The Bayesian fit is cached with the generated application data; setting `TRUNCCOMP_REFRESH_IST3_BAYES=true` regenerates it from the fixed-width file. Because this Bayesian implementation uses a practical finite truncation for the manuscript build, we report sampler and truncation diagnostics alongside the posterior summaries.

The posterior summaries in Table 5 separate the survival/atom component from the survivor EQ-

Table 5: Bayesian IST-3 posterior summaries from the two-part mixture model.

Quantity	Posterior median	95% CrI	Probability
Death probability, control	0.299	0.28 to 0.32	
Death probability, rt-PA	0.297	0.27 to 0.32	
Alive/EQ-VAS probability, control	0.701	0.68 to 0.72	
Alive/EQ-VAS probability, rt-PA	0.703	0.68 to 0.73	
Survivor mean EQ-VAS, control	57.89	56.45 to 59.31	
Survivor mean EQ-VAS, rt-PA	60.70	59.22 to 62.24	
Death risk difference	-0.003	-0.03 to 0.03	$\Pr(< 0) = 0.579$
μ_δ	2.82	0.71 to 4.98	$\Pr(> 0) = 0.997$
α_δ	1.015	0.86 to 1.17	$\Pr(> 1) = 0.579$
Δ	2.14	-0.30 to 4.52	$\Pr(> 0) = 0.955$
Sampler diagnostics	–	–	Divergences 0; max $\hat{R}$ 1.014; min bulk ESS 872
Mixture truncation check	–	–	Tail q95 0.541/0.464; passed = no

VAS component. The benefit-oriented combined endpoint uses the same coding as the frequentist analysis, so $\Pr(\Delta > 0 \mid \text{data})$ is the posterior probability that rt-PA improves the combined endpoint. The posterior probabilities $\Pr(\mu_\delta > 0 \mid \text{data})$ and $\Pr(\alpha_\delta > 1 \mid \text{data})$ separately summarize benefit on survivor EQ-VAS and on avoiding the atom. Figure 4 shows the posterior densities for these component and combined contrasts.

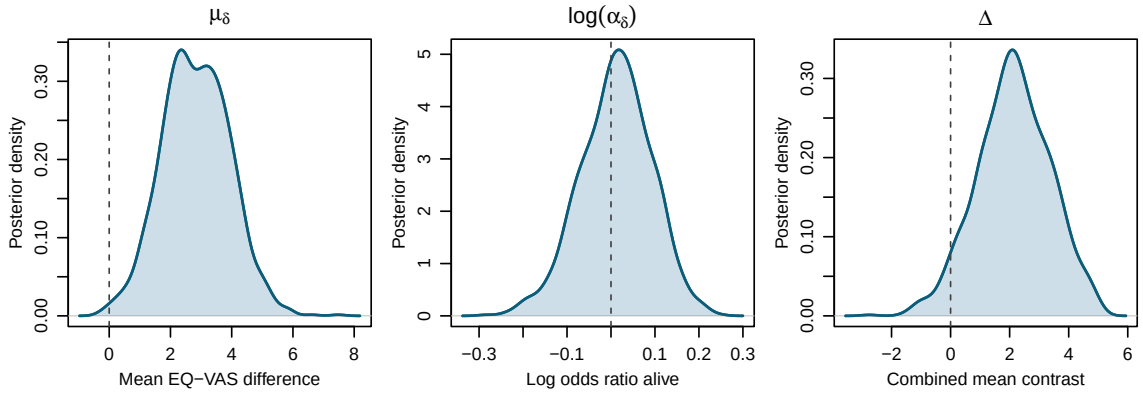

Figure 4: Posterior distributions for the IST-3 treatment contrasts. Vertical dashed lines mark no survivor mean difference, no difference in the odds of being alive with observed EQ-VAS, and no combined endpoint mean difference.

Finally, we assessed whether the Bayesian model reproduced both parts of the endpoint. Table 6 reports posterior predictive checks for the atom model and the survivor EQ-VAS distribution, and Figure 5 displays the corresponding replicated data summaries. These checks are important because a model can match the death proportions while still misrepresenting the bounded and potentially skewed EQ-VAS distribution among survivors. In this application the atom component is reproduced well, whereas the continuous-component check is deliberately treated as a warning about the simple real-line mixture approximation to a bounded health-state score. The posterior predictive assessment is therefore used as a model check, not as evidence for treatment efficacy.

Table 6: Posterior predictive checks for the IST-3 Bayesian model.

Component	Statistic	Posterior predictive p	Notes
Death atom	Maximum armwise absolute atom-rate deviation	0.515	200 draws; Indicator for Y = atom scale
Survivor EQ-VAS	Maximum armwise KS discrepancy	< 0.001	200 draws; Outcome scale

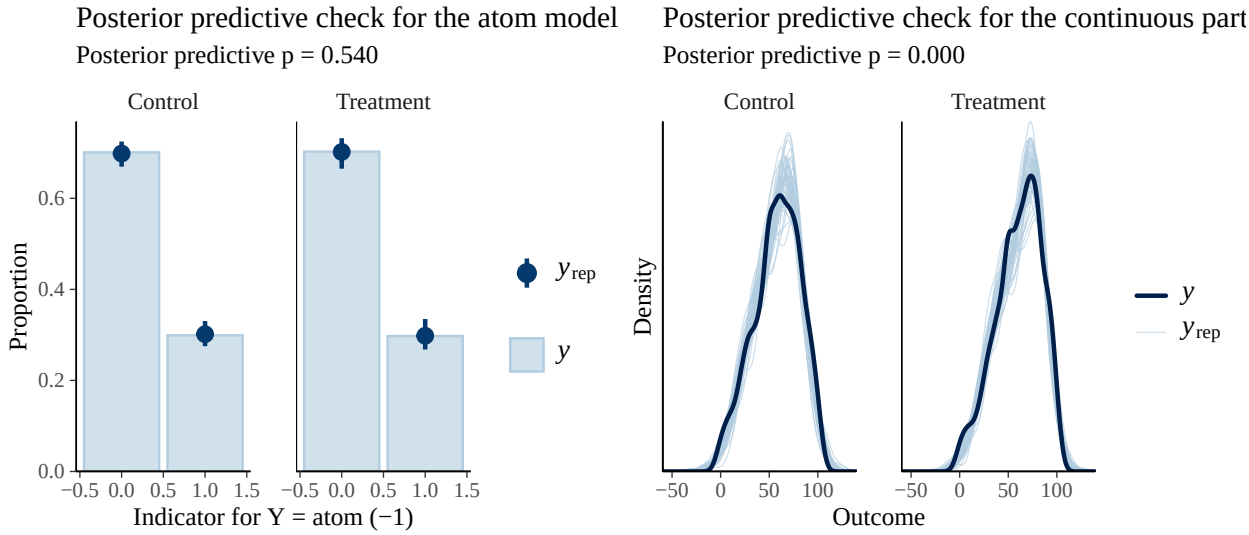

Figure 5: Posterior predictive checks for the IST-3 Bayesian model, separately for the death atom and the survivor EQ-VAS component.

5 Discussion

We have proposed parametric and semi-parametric likelihood-ratio tests for continuous outcomes with a clinically meaningful atom. The key step is to retain the atomized outcome used in practice while testing the joint null of no treatment effect on the atom probability and no treatment effect on the positive part. This yields one prespecified p-value together with interpretable component-wise effect estimates and confidence regions.

The simulations show why that distinction matters. The gain is largest when treatment acts mainly through one component or when the atom and positive part move in opposite directions, because those are exactly the settings in which a one-dimensional composite can dilute or cancel signal. The IST-3 application serves the same purpose in a clinically recognizable setting: the joint tests detect two-component structure that is only weakly visible to standard analyses of the raw combined outcome.

This observed-data focus should also guide interpretation. The parameter pair $(\mu_\delta, \beta_\delta)$ is not a survivor-average causal effect or a principal-stratum estimand; it describes the two components of the observed outcome distribution under randomization. The procedures are asymptotic, require positive numbers of observed outcomes in both groups, and the parametric version is only as credible as the model specified for $Y \mid A = 1$. The semi-parametric version relaxes that distributional assumption, but it still relies on regularity of the Bernoulli model and on feasible empirical-likelihood constraints.

For trials that require a single prespecified primary analysis, the framework offers one joint p-value together with clinically interpretable follow-up summaries for the binary and continuous components. That makes it attractive when investigators want to preserve the meaning of a worst-rank or atom-at-zero endpoint without collapsing the scientific question to a single distribution on the combined scale.

Software implementing both procedures, including the code used to generate the manuscript analyses, is available on GitHub (Jensen and Lange, 2018).

A Supplementary simulation results

This appendix collects the simulation figures and tables referenced in Section 3. It also records the exact simulation protocol implemented in `simulation-study/scripts/run-simulation-study.R` and `simulation-study/R/simulation-study.R`.

The simulation study used the full factorial design generated by the code defaults: six scenarios, four effect levels $h \in \{0, 1, 2, 3\}$, and ten per-group sample sizes $n \in \{50, 100, \ldots, 500\}$, for a total of 240 cells. Each replicate contained exactly n control observations and n treatment observations. Within group $r \in \{0, 1\}$, the indicator A of whether the continuous outcome was observed was drawn as $A \sim \text{Bernoulli}(\pi_r)$. Conditional on $A = 1$, the positive outcome was drawn from a scenario-specific distribution $F_{r,h}$; conditional on $A = 0$, the numerical atom value was recorded as 0. Thus the analyzed endpoint was always the atom-at-zero numerical combined outcome $\widetilde{Y}^0 = AY$, equivalently $c_0(\widetilde{Y})$, and no covariate adjustment was used.

The six scenario families were as follows. Scenario 1 used $\pi_0 = \pi_1 = 0.85$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(3 + \delta_h, 1)$ with $\delta_h \in \{0, 0.20, 0.35, 0.50\}$. Scenario 2 used $\pi_0 = 0.45$, $\pi_1 \in \{0.45, 0.50, 0.55, 0.60\}$, and $Y \mid A = 1, R = 0 \sim Y \mid A = 1, R = 1 \sim N(3, 1)$. Scenario 3 used $\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(1.5/\pi_1, 1)$, so that $E(\widetilde{Y}^0 \mid R = 1) = E(\widetilde{Y}^0 \mid R = 0) = 1.5$ at every effect level even though both components change. Scenario 4 used $\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(3 + 0.15h, 1)$. Scenario 5 used $\pi_0 = \pi_1 = 0.65$, $Y \mid A = 1, R = 0 \sim \Gamma(9, 3/9)$, and $Y \mid A = 1, R = 1 \sim \Gamma(k_h, m_h/k_h)$ with $(k_h, m_h) \in \{(9, 3.0), (6, 3.2), (4, 3.4), (2.5, 3.6)\}$, where $\Gamma(k, \theta)$ denotes a gamma distribution with shape k and scale θ. Scenario 6 used $\pi_0 = \pi_1 = 0.80$ and $Y \mid A = 1, R = r \sim 0.95\, N(3 + \delta_h r, 0.35^2) + 0.05\, N(10 + \delta_h r, 0.5^2)$ with $\delta_h \in \{0, 0.10, 0.20, 0.35\}$.

For each cell, the driver called `set.seed(20260416 + cell_id - 1)` before generating the 10,000 replicates, where `cell_id` increases first over scenarios, then over effect levels, and then over sample sizes. Each replicate was analyzed by four methods applied to the combined outcome data frame produced by `simulateTruncatedData()`: `wilcox.test(Y ~ R, exact = FALSE)` for the Wilcoxon rank-sum test, `t.test(Y ~ R)` for Welch's two-sample t-test, `truncComp(Y ~ R, atom = 0, method = "LRT")` for the parametric likelihood-ratio test, and `truncComp(Y ~ R, atom = 0, method = "SPLRT")` for the semi-parametric likelihood-ratio test. In the package implementation, the parametric test combines a logistic likelihood-ratio comparison for A with a Gaussian likelihood-ratio comparison for Y among observations with $A = 1$, whereas the semi-parametric test combines the same logistic component with an empirical-likelihood test for the survivor mean difference. Both proposed procedures report a two-degree-of-freedom χ^2-calibrated joint p-value.

Within each cell, rejection rates were computed as the proportion of successful replicates with p-value below 0.05 for each method, and the Monte Carlo standard error was computed from that rejection proportion using the number of successful replicates for the given method. If a method call errored or returned a non-finite p-value, that replicate was counted as a failure for that method. The aggregated results object stores method-specific failure rates and runtimes, and the supplementary tables report the largest failure rate across the four methods in each cell. For the manuscript displays, Figure 1 uses the strongest non-null effect level ($h = 3$) across all six scenarios; Figure 2 uses all four effect levels at $n = 250$ per group across all six scenarios; and Figure 6 uses the null cells with $h = 0$ across all six scenarios. The six long supplementary tables list the complete power results for all 240 cells together with the expected coded combined-outcome

contrast $E(\widetilde{Y}^0 \mid R = 1) - E(\widetilde{Y}^0 \mid R = 0)$ used in the code-generated summaries.

Figure 6 summarizes the $h = 0$ null-calibration results across all six scenarios.

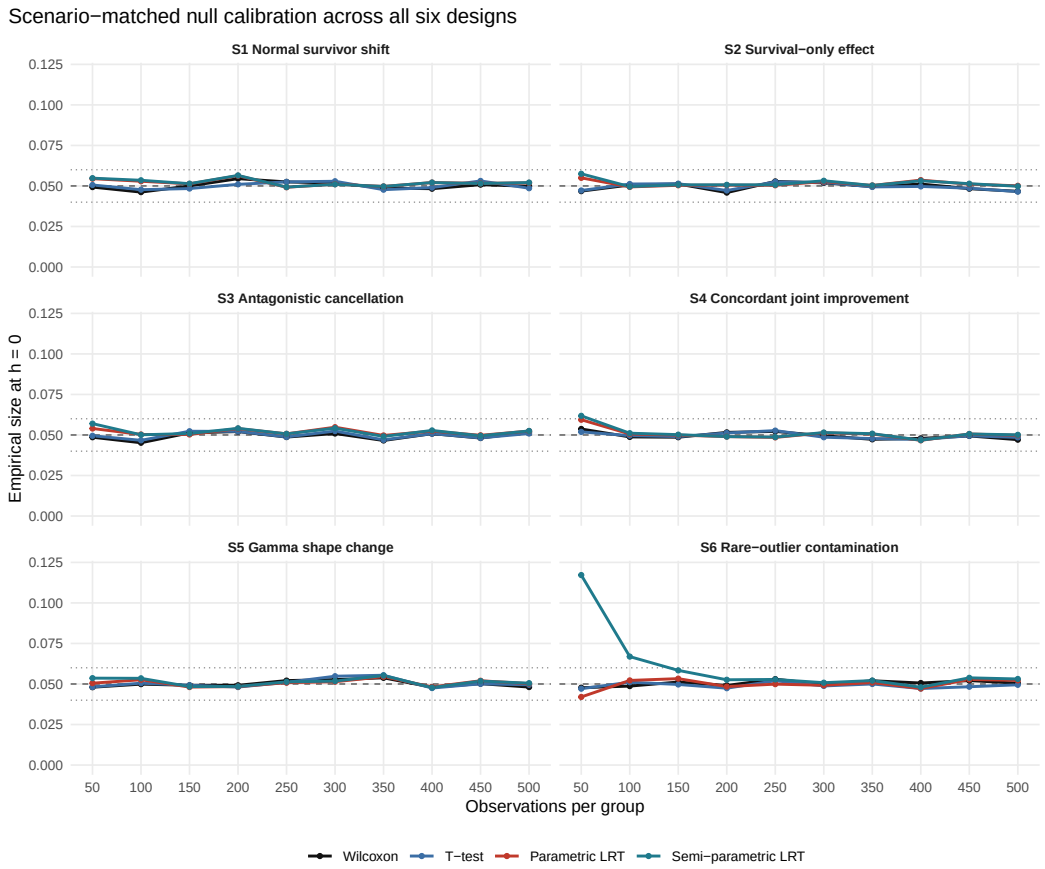

Figure 6: Empirical size at effect level $h = 0$ across all six simulation scenarios. The dashed line marks 0.05 and the dotted lines mark 0.04 and 0.06.

Table 7: Normal survivor shift, low atom: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\delta = 0.00$	0.000	4.93	5.06	5.45	5.48	0
100	0	$\delta = 0.00$	0.000	4.62	4.77	5.29	5.35	0
150	0	$\delta = 0.00$	0.000	5.00	4.84	5.13	5.14	0
200	0	$\delta = 0.00$	0.000	5.45	5.10	5.63	5.65	0
250	0	$\delta = 0.00$	0.000	5.25	5.25	4.92	4.93	0
300	0	$\delta = 0.00$	0.000	5.18	5.29	5.11	5.10	0
350	0	$\delta = 0.00$	0.000	4.87	4.77	4.97	4.97	0
400	0	$\delta = 0.00$	0.000	4.83	4.90	5.21	5.21	0
450	0	$\delta = 0.00$	0.000	5.08	5.31	5.16	5.15	0
500	0	$\delta = 0.00$	0.000	4.98	4.86	5.19	5.21	0
50	1	$\delta = 0.20$	0.170	10.70	8.87	12.49	12.37	0
100	1	$\delta = 0.20$	0.170	16.25	12.83	19.60	19.62	0

Table 7: Normal survivor shift, low atom: estimated power by sample size and effect level *(continued)*

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
150	1	$\delta = 0.20$	0.170	22.82	17.65	28.49	28.54	0
200	1	$\delta = 0.20$	0.170	29.59	22.23	36.83	36.96	0
250	1	$\delta = 0.20$	0.170	34.79	25.97	43.79	43.86	0
300	1	$\delta = 0.20$	0.170	40.67	30.31	50.56	50.50	0
350	1	$\delta = 0.20$	0.170	46.48	34.71	58.72	58.73	0
400	1	$\delta = 0.20$	0.170	51.52	38.62	64.27	64.32	0
450	1	$\delta = 0.20$	0.170	56.24	41.90	69.66	69.73	0
500	1	$\delta = 0.20$	0.170	60.71	46.75	74.71	74.68	0
50	2	$\delta = 0.35$	0.298	23.18	17.79	29.19	29.45	0
100	2	$\delta = 0.35$	0.298	41.40	30.40	52.87	52.94	0
150	2	$\delta = 0.35$	0.298	57.19	42.24	71.19	71.14	0
200	2	$\delta = 0.35$	0.298	69.15	52.95	82.93	82.95	0
250	2	$\delta = 0.35$	0.298	77.50	60.74	90.65	90.65	0
300	2	$\delta = 0.35$	0.298	85.03	70.09	95.15	95.12	0
350	2	$\delta = 0.35$	0.298	90.28	76.72	97.40	97.38	0
400	2	$\delta = 0.35$	0.298	93.29	81.88	98.79	98.80	0
450	2	$\delta = 0.35$	0.298	95.97	85.93	99.39	99.38	0
500	2	$\delta = 0.35$	0.298	97.25	89.24	99.69	99.68	0
50	3	$\delta = 0.50$	0.425	42.16	31.65	54.08	54.38	0
100	3	$\delta = 0.50$	0.425	69.07	51.76	83.35	83.50	0
150	3	$\delta = 0.50$	0.425	85.75	69.34	95.55	95.54	0
200	3	$\delta = 0.50$	0.425	93.70	80.90	99.03	99.05	0
250	3	$\delta = 0.50$	0.425	97.46	89.03	99.76	99.77	0
300	3	$\delta = 0.50$	0.425	98.97	93.05	99.95	99.95	0
350	3	$\delta = 0.50$	0.425	99.66	96.51	100.00	100.00	0
400	3	$\delta = 0.50$	0.425	99.86	97.78	100.00	100.00	0
450	3	$\delta = 0.50$	0.425	99.92	99.04	100.00	100.00	0
500	3	$\delta = 0.50$	0.425	99.97	99.53	100.00	100.00	0

Table 8: Survival-only effect: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\pi_1 = 0.45$	0.00	4.68	4.72	5.50	5.75	0
100	0	$\pi_1 = 0.45$	0.00	5.06	5.12	4.94	4.96	0
150	0	$\pi_1 = 0.45$	0.00	5.12	5.14	5.04	5.07	0
200	0	$\pi_1 = 0.45$	0.00	4.59	4.72	5.05	5.07	0
250	0	$\pi_1 = 0.45$	0.00	5.28	5.25	5.04	5.07	0
300	0	$\pi_1 = 0.45$	0.00	5.19	5.22	5.27	5.32	0
350	0	$\pi_1 = 0.45$	0.00	4.96	4.94	5.03	5.03	0
400	0	$\pi_1 = 0.45$	0.00	5.12	4.97	5.36	5.32	0
450	0	$\pi_1 = 0.45$	0.00	4.83	4.86	5.11	5.14	0
500	0	$\pi_1 = 0.45$	0.00	4.67	4.65	5.01	4.98	0

Table 8: Survival-only effect: estimated power by sample size and effect level *(continued)*

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	1	$\pi_1 = 0.50$	0.15	7.31	7.28	7.36	7.63	0
100	1	$\pi_1 = 0.50$	0.15	9.79	9.55	9.32	9.38	0
150	1	$\pi_1 = 0.50$	0.15	12.62	12.27	11.06	11.14	0
200	1	$\pi_1 = 0.50$	0.15	15.53	15.11	13.90	13.88	0
250	1	$\pi_1 = 0.50$	0.15	18.50	17.88	15.71	15.70	0
300	1	$\pi_1 = 0.50$	0.15	20.48	19.79	17.84	17.83	0
350	1	$\pi_1 = 0.50$	0.15	23.59	22.86	20.43	20.44	0
400	1	$\pi_1 = 0.50$	0.15	26.17	25.10	22.72	22.74	0
450	1	$\pi_1 = 0.50$	0.15	29.23	28.19	25.74	25.74	0
500	1	$\pi_1 = 0.50$	0.15	30.79	29.86	27.05	27.08	0
50	2	$\pi_1 = 0.55$	0.30	14.71	14.59	13.41	13.65	0
100	2	$\pi_1 = 0.55$	0.30	25.71	24.88	22.57	22.72	0
150	2	$\pi_1 = 0.55$	0.30	35.50	34.57	32.44	32.40	0
200	2	$\pi_1 = 0.55$	0.30	45.04	43.39	41.36	41.36	0
250	2	$\pi_1 = 0.55$	0.30	54.45	52.87	50.16	50.11	0
300	2	$\pi_1 = 0.55$	0.30	62.31	60.34	59.07	59.04	0
350	2	$\pi_1 = 0.55$	0.30	69.18	67.67	66.14	66.16	0
400	2	$\pi_1 = 0.55$	0.30	74.67	72.88	72.27	72.29	0
450	2	$\pi_1 = 0.55$	0.30	79.61	77.82	77.85	77.85	0
500	2	$\pi_1 = 0.55$	0.30	82.89	81.21	81.91	81.92	0
50	3	$\pi_1 = 0.60$	0.45	27.77	27.73	25.77	25.97	0
100	3	$\pi_1 = 0.60$	0.45	49.82	48.92	47.52	47.52	0
150	3	$\pi_1 = 0.60$	0.45	65.55	64.81	64.34	64.30	0
200	3	$\pi_1 = 0.60$	0.45	79.20	78.09	77.46	77.49	0
250	3	$\pi_1 = 0.60$	0.45	87.01	86.32	86.71	86.73	0
300	3	$\pi_1 = 0.60$	0.45	92.19	91.45	92.00	91.99	0
350	3	$\pi_1 = 0.60$	0.45	95.02	94.64	95.65	95.64	0
400	3	$\pi_1 = 0.60$	0.45	97.32	96.96	97.59	97.59	0
450	3	$\pi_1 = 0.60$	0.45	98.47	98.31	98.77	98.77	0
500	3	$\pi_1 = 0.60$	0.45	99.08	98.96	99.33	99.33	0

Table 9: Antagonistic cancellation: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	4.86	4.95	5.40	5.70	0
100	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	4.52	4.67	5.03	5.01	0
150	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	5.13	5.23	5.03	5.09	0
200	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	5.21	5.22	5.39	5.41	0
250	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	4.87	4.88	5.07	5.07	0
300	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	5.09	5.24	5.48	5.42	0
350	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	4.66	4.70	4.97	4.91	0
400	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	5.08	5.08	5.24	5.28	0
450	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	4.81	4.82	4.98	4.95	0

Table 9: Antagonistic cancellation: estimated power by sample size and effect level *(continued)*

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
500	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.000$	0	5.18	5.09	5.24	5.25	0
50	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	4.80	4.82	15.91	16.20	0
100	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.14	5.01	27.50	27.71	0
150	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.29	5.07	38.21	38.20	0
200	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.59	5.51	49.94	50.03	0
250	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.51	5.20	58.85	58.85	0
300	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.03	4.65	67.66	67.76	0
350	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.17	5.01	74.96	75.03	0
400	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.72	5.31	80.17	80.15	0
450	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	4.91	4.65	84.97	84.96	0
500	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 2.727$	0	5.74	5.25	88.89	88.91	0
50	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	5.09	4.83	44.85	45.24	0
100	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	5.55	5.30	76.11	76.19	0
150	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	5.96	4.98	91.29	91.28	0
200	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	6.19	5.06	97.43	97.38	0
250	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	5.92	4.82	99.34	99.34	0
300	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	6.34	4.89	99.73	99.74	0
350	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	6.71	5.20	99.93	99.93	0
400	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	6.22	4.83	100.00	100.00	0
450	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	6.57	4.99	100.00	100.00	0
500	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 2.500$	0	7.30	5.18	100.00	100.00	0
50	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	6.03	5.32	78.29	78.70	0
100	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	6.25	4.99	97.70	97.74	0
150	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	6.68	4.67	99.73	99.73	0
200	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	7.08	5.00	99.99	99.99	0
250	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	7.84	4.83	100.00	100.00	0
300	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	8.45	5.12	100.00	100.00	0
350	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	8.14	4.78	100.00	100.00	0
400	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	9.07	5.32	100.00	100.00	0
450	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	9.40	5.20	100.00	100.00	0
500	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 2.308$	0	10.32	5.10	100.00	100.00	0

Table 10: Concordant joint improvement: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	5.37	5.19	5.94	6.18	0
100	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	4.88	4.96	5.06	5.11	0
150	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	4.86	4.90	4.99	5.02	0
200	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	5.16	5.12	4.90	4.90	0
250	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	5.22	5.27	4.85	4.87	0
300	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	4.95	4.86	5.13	5.15	0
350	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	4.73	4.78	5.06	5.08	0
400	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	4.79	4.72	4.67	4.68	0

Table 10: Concordant joint improvement: estimated power by sample size and effect level *(continued)*

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
450	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	4.94	4.95	5.06	5.06	0
500	0	$\pi_1 = 0.50$, $\mu_0 + \mu_\delta = 3.00$	0.000	4.71	4.85	4.98	5.01	0
50	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	10.04	10.43	9.94	10.07	0
100	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	16.77	17.12	14.71	14.81	0
150	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	21.12	21.79	19.35	19.41	0
200	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	26.60	27.35	24.55	24.47	0
250	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	32.82	33.75	29.38	29.44	0
300	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	37.94	38.66	34.56	34.64	0
350	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	42.92	44.26	40.05	40.09	0
400	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	47.96	48.91	45.25	45.23	0
450	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	52.00	53.26	48.90	48.97	0
500	1	$\pi_1 = 0.55$, $\mu_0 + \mu_\delta = 3.15$	0.233	56.58	57.97	54.48	54.42	0
50	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	26.99	27.49	25.21	25.63	0
100	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	49.75	50.61	47.03	46.96	0
150	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	65.65	66.19	63.14	63.17	0
200	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	78.17	78.70	76.24	76.19	0
250	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	86.73	87.11	86.21	86.22	0
300	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	92.43	92.74	92.36	92.32	0
350	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	95.42	95.70	95.45	95.45	0
400	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	97.43	97.48	97.29	97.31	0
450	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	98.47	98.60	98.63	98.63	0
500	2	$\pi_1 = 0.60$, $\mu_0 + \mu_\delta = 3.30$	0.480	99.15	99.25	99.22	99.21	0
50	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	55.74	55.95	53.18	53.60	0
100	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	84.80	84.63	83.98	84.05	0
150	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	95.59	95.53	95.24	95.20	0
200	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	98.92	98.93	98.95	98.94	0
250	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	99.65	99.65	99.68	99.67	0
300	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	99.97	99.97	99.94	99.94	0
350	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	99.99	99.98	99.98	99.98	0
400	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	100.00	100.00	100.00	100.00	0
450	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	100.00	100.00	100.00	100.00	0
500	3	$\pi_1 = 0.65$, $\mu_0 + \mu_\delta = 3.45$	0.743	100.00	100.00	100.00	100.00	0

Table 11: Gamma shape-change with mean shift: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	4.80	4.82	5.05	5.36	0
100	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	4.99	5.05	5.26	5.35	0
150	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	4.92	4.92	4.81	4.84	0
200	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	4.92	4.81	4.84	4.85	0
250	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	5.21	5.10	5.08	5.13	0
300	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	5.27	5.48	5.14	5.18	0

Table 11: Gamma shape-change with mean shift: estimated power by sample size and effect level *(continued)*

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
350	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	5.38	5.54	5.45	5.52	0
400	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	4.80	4.75	4.82	4.78	0
450	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	5.02	5.01	5.20	5.18	0
500	0	$k_1 = 9.0$, $\mu_0 + \mu_\delta = 3.0$	0.00	4.81	4.95	5.02	5.05	0
50	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	5.75	6.39	8.19	9.09	0
100	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	6.60	8.15	13.01	13.70	0
150	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	7.42	9.99	17.43	18.17	0
200	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	7.24	10.47	20.44	21.32	0
250	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	8.50	12.57	25.74	26.62	0
300	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	9.25	14.51	30.82	31.63	0
350	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	10.06	16.17	35.00	35.90	0
400	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	11.35	18.41	39.75	40.59	0
450	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	11.81	19.46	43.84	44.59	0
500	1	$k_1 = 6.0$, $\mu_0 + \mu_\delta = 3.2$	0.13	12.85	21.94	49.29	50.23	0
50	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	6.87	10.16	15.58	18.71	0
100	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	8.05	15.12	27.88	30.82	0
150	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	10.30	21.32	40.92	44.36	0
200	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	11.45	26.18	52.58	55.33	0
250	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	13.48	33.01	63.64	66.09	0
300	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	14.73	38.39	72.47	74.39	0
350	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	16.48	42.83	79.05	80.67	0
400	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	18.86	49.48	85.01	86.15	0
450	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	19.95	54.08	89.59	90.50	0
500	2	$k_1 = 4.0$, $\mu_0 + \mu_\delta = 3.4$	0.26	21.98	58.84	92.32	93.03	0
50	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	6.03	12.97	19.07	24.98	0
100	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	7.61	23.80	37.88	44.38	0
150	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	9.00	33.78	55.67	61.08	0
200	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	10.01	44.34	70.14	74.17	0
250	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	11.79	53.43	81.37	84.28	0
300	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	12.67	61.52	87.98	90.04	0
350	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	13.83	68.20	92.98	94.26	0
400	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	15.38	73.69	95.73	96.53	0
450	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	16.38	79.98	97.57	97.98	0
500	3	$k_1 = 2.5$, $\mu_0 + \mu_\delta = 3.6$	0.39	17.99	83.62	98.59	98.85	0

Table 12: Rare-outlier contamination: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
50	0	$\delta = 0.00$	0.00	4.77	4.71	4.20	11.72	0
100	0	$\delta = 0.00$	0.00	4.87	5.11	5.22	6.69	0
150	0	$\delta = 0.00$	0.00	5.12	4.97	5.33	5.84	0
200	0	$\delta = 0.00$	0.00	4.91	4.74	4.84	5.26	0

Table 12: Rare-outlier contamination: estimated power by sample size and effect level *(continued)*

n	h	Effect	Expected Δ	Wilcoxon	T-test	LRT	SPLRT	Max fail
250	0	$\delta = 0.00$	0.00	5.30	5.20	4.99	5.28	0
300	0	$\delta = 0.00$	0.00	4.96	4.88	4.92	5.08	0
350	0	$\delta = 0.00$	0.00	5.20	5.00	5.09	5.21	0
400	0	$\delta = 0.00$	0.00	5.06	4.72	4.72	4.80	0
450	0	$\delta = 0.00$	0.00	5.21	4.83	5.28	5.38	0
500	0	$\delta = 0.00$	0.00	5.05	4.95	5.22	5.31	0
50	1	$\delta = 0.10$	0.08	12.06	5.15	5.94	13.39	0
100	1	$\delta = 0.10$	0.08	20.46	5.80	6.41	7.38	0
150	1	$\delta = 0.10$	0.08	28.91	6.56	6.92	7.15	0
200	1	$\delta = 0.10$	0.08	36.21	6.84	7.46	7.48	0
250	1	$\delta = 0.10$	0.08	43.78	7.50	8.78	8.79	0
300	1	$\delta = 0.10$	0.08	50.91	8.15	9.27	9.20	0
350	1	$\delta = 0.10$	0.08	58.09	8.93	9.86	9.86	0
400	1	$\delta = 0.10$	0.08	62.34	9.17	10.93	10.82	0
450	1	$\delta = 0.10$	0.08	67.37	9.04	10.60	10.50	0
500	1	$\delta = 0.10$	0.08	71.56	9.77	11.99	11.85	0
50	2	$\delta = 0.20$	0.16	35.50	7.27	9.67	15.57	0
100	2	$\delta = 0.20$	0.16	61.02	8.95	11.76	11.68	0
150	2	$\delta = 0.20$	0.16	77.92	10.34	13.60	12.98	0
200	2	$\delta = 0.20$	0.16	87.72	12.72	16.64	15.81	0
250	2	$\delta = 0.20$	0.16	94.23	14.63	20.15	19.16	0
300	2	$\delta = 0.20$	0.16	97.38	16.46	22.81	21.78	0
350	2	$\delta = 0.20$	0.16	98.61	19.85	26.33	25.10	0
400	2	$\delta = 0.20$	0.16	99.39	22.00	29.88	28.58	0
450	2	$\delta = 0.20$	0.16	99.67	22.89	32.64	31.42	0
500	2	$\delta = 0.20$	0.16	99.93	25.55	36.22	34.88	0
50	3	$\delta = 0.35$	0.28	74.80	11.59	18.33	18.94	0
100	3	$\delta = 0.35$	0.28	96.29	17.18	25.12	21.95	0
150	3	$\delta = 0.35$	0.28	99.59	23.77	34.81	31.34	0
200	3	$\delta = 0.35$	0.28	99.93	30.07	43.19	39.66	0
250	3	$\delta = 0.35$	0.28	99.99	35.51	51.19	47.93	0
300	3	$\delta = 0.35$	0.28	100.00	39.96	58.62	55.31	0
350	3	$\delta = 0.35$	0.28	100.00	46.09	65.06	62.04	0
400	3	$\delta = 0.35$	0.28	100.00	51.28	72.13	69.67	0
450	3	$\delta = 0.35$	0.28	100.00	56.17	77.17	75.01	0
500	3	$\delta = 0.35$	0.28	100.00	60.66	81.06	79.38	0

Table 13: Scenario-matched null calibration at effect level $h = 0$

Scenario	n	Wilcoxon	T-test	LRT	SPLRT	Max fail
S1	50	4.93	5.06	5.45	5.48	0
S1	100	4.62	4.77	5.29	5.35	0
S1	150	5.00	4.84	5.13	5.14	0

Table 13: Scenario-matched null calibration at effect level $h = 0$ *(continued)*

Scenario	n	Wilcoxon	T-test	LRT	SPLRT	Max fail
S1	200	5.45	5.10	5.63	5.65	0
S1	250	5.25	5.25	4.92	4.93	0
S1	300	5.18	5.29	5.11	5.10	0
S1	350	4.87	4.77	4.97	4.97	0
S1	400	4.83	4.90	5.21	5.21	0
S1	450	5.08	5.31	5.16	5.15	0
S1	500	4.98	4.86	5.19	5.21	0
S2	50	4.68	4.72	5.50	5.75	0
S2	100	5.06	5.12	4.94	4.96	0
S2	150	5.12	5.14	5.04	5.07	0
S2	200	4.59	4.72	5.05	5.07	0
S2	250	5.28	5.25	5.04	5.07	0
S2	300	5.19	5.22	5.27	5.32	0
S2	350	4.96	4.94	5.03	5.03	0
S2	400	5.12	4.97	5.36	5.32	0
S2	450	4.83	4.86	5.11	5.14	0
S2	500	4.67	4.65	5.01	4.98	0
S3	50	4.86	4.95	5.40	5.70	0
S3	100	4.52	4.67	5.03	5.01	0
S3	150	5.13	5.23	5.03	5.09	0
S3	200	5.21	5.22	5.39	5.41	0
S3	250	4.87	4.88	5.07	5.07	0
S3	300	5.09	5.24	5.48	5.42	0
S3	350	4.66	4.70	4.97	4.91	0
S3	400	5.08	5.08	5.24	5.28	0
S3	450	4.81	4.82	4.98	4.95	0
S3	500	5.18	5.09	5.24	5.25	0
S4	50	5.37	5.19	5.94	6.18	0
S4	100	4.88	4.96	5.06	5.11	0
S4	150	4.86	4.90	4.99	5.02	0
S4	200	5.16	5.12	4.90	4.90	0
S4	250	5.22	5.27	4.85	4.87	0
S4	300	4.95	4.86	5.13	5.15	0
S4	350	4.73	4.78	5.06	5.08	0
S4	400	4.79	4.72	4.67	4.68	0
S4	450	4.94	4.95	5.06	5.06	0
S4	500	4.71	4.85	4.98	5.01	0
S5	50	4.80	4.82	5.05	5.36	0
S5	100	4.99	5.05	5.26	5.35	0
S5	150	4.92	4.92	4.81	4.84	0
S5	200	4.92	4.81	4.84	4.85	0
S5	250	5.21	5.10	5.08	5.13	0
S5	300	5.27	5.48	5.14	5.18	0
S5	350	5.38	5.54	5.45	5.52	0

Table 13: Scenario-matched null calibration at effect level $h = 0$ *(continued)*

Scenario	n	Wilcoxon	T-test	LRT	SPLRT	Max fail
S5	400	4.80	4.75	4.82	4.78	0
S5	450	5.02	5.01	5.20	5.18	0
S5	500	4.81	4.95	5.02	5.05	0
S6	50	4.77	4.71	4.20	11.72	0
S6	100	4.87	5.11	5.22	6.69	0
S6	150	5.12	4.97	5.33	5.84	0
S6	200	4.91	4.74	4.84	5.26	0
S6	250	5.30	5.20	4.99	5.28	0
S6	300	4.96	4.88	4.92	5.08	0
S6	350	5.20	5.00	5.09	5.21	0
S6	400	5.06	4.72	4.72	4.80	0
S6	450	5.21	4.83	5.28	5.38	0
S6	500	5.05	4.95	5.22	5.31	0

B Supplementary description of the Delta-scale profile and projected intervals

This appendix records the exact unadjusted `TruncComp2` constructions underlying `DeltaProfileCI` and `DeltaProjectedCI`. These intervals are currently implemented only for the no-covariate parametric and semi-parametric models described in Section 2. To unify notation, define

$$\mathcal{W}_n(\mu_\delta, \beta_\delta) = \begin{cases} W_n^P(\mu_\delta, \beta_\delta), & \text{for the parametric fit,} \\ \widetilde{W}_n^P(\mu_\delta, \beta_\delta), & \text{for the semi-parametric fit.} \end{cases}$$

Thus $\mathcal{W}_n$ always denotes the fitted two-parameter joint likelihood-ratio surface on the $(\mu_\delta, \beta_\delta)$ scale.

For the binary component, let

$$\ell_{A,n}(\beta_0, \beta_\delta) = \sum_{i=1}^{n} \left\{ \log\left(1 + e^{\beta_0 + \beta_\delta R_i}\right) - A_i(\beta_0 + \beta_\delta R_i) \right\}$$

denote the negative Bernoulli log-likelihood. For fixed β_δ, define the profiled intercept

$$\widehat{\beta}_0(\beta_\delta) \in \arg\min_{\beta_0 \in \mathbb{R}} \ell_{A,n}(\beta_0, \beta_\delta),$$

and the corresponding fitted observation probabilities

$$\widehat{\pi}_r(\beta_\delta) = \text{logit}^{-1}\left(\widehat{\beta}_0(\beta_\delta) + r\beta_\delta\right), \qquad r \in \{0, 1\}.$$

For the continuous component, let $\widehat{\mu}_0(\mu_\delta)$ denote the profiled control-group mean among the observed outcomes under the fixed mean-difference constraint μ_δ:

$$\widehat{\mu}_0(\mu_\delta) \in \begin{cases} \arg\max_{\mu_0, \Psi} L_{Y,n}(\mu_0, \mu_\delta, \Psi), & \text{parametric,} \\ \arg\min_{\mu_0 \in \mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, \mu_\delta), & \text{semi-parametric,} \end{cases}$$

and define $\widehat{\mu}_1(\mu_\delta) = \widehat{\mu}_0(\mu_\delta) + \mu_\delta$. The profiled atom-at-zero numerical combined-outcome contrast induced by the pair $(\mu_\delta, \beta_\delta)$ is then

$$g_n(\mu_\delta, \beta_\delta) = \widehat{\pi}_1(\beta_\delta)\widehat{\mu}_1(\mu_\delta) - \widehat{\pi}_0(\beta_\delta)\widehat{\mu}_0(\mu_\delta). \tag{30}$$

Equation (30) is the map that the software evaluates at every joint candidate point on the atom-at-zero numerical combined-outcome scale. In particular, the reported point estimate `Delta` is $\widehat{\Delta}_n := g_n(\widehat{\mu}_\delta, \widehat{\beta}_\delta)$.

Let

$$c_{1,\alpha} = \chi^2_1(1-\alpha), \qquad c_{2,\alpha} = \chi^2_2(1-\alpha)$$

denote the one- and two-degree-of-freedom upper quantiles used for the profile and projected intervals on the atom-at-zero numerical combined-outcome scale Δ, respectively.

Projected interval

The projected interval is the image of the two-dimensional simultaneous confidence region under the map in Equation (30). More precisely, the package reports the endpoint pair

$$\mathrm{CI}^{\Delta,\mathrm{proj}}_{1-\alpha} = \left[L^{\mathrm{proj}}_{1-\alpha}, U^{\mathrm{proj}}_{1-\alpha}\right],$$

where

$$L^{\mathrm{proj}}_{1-\alpha} = \inf\left\{g_n(\mu_\delta, \beta_\delta) : \mathcal{W}_n(\mu_\delta, \beta_\delta) \le c_{2,\alpha}\right\}, \tag{31}$$

$$U^{\mathrm{proj}}_{1-\alpha} = \sup\left\{g_n(\mu_\delta, \beta_\delta) : \mathcal{W}_n(\mu_\delta, \beta_\delta) \le c_{2,\alpha}\right\}. \tag{32}$$

Thus the projected interval is the smallest closed interval containing all Δ values induced by joint parameter pairs inside the $(1-\alpha)100\%$ simultaneous region for $(\mu_\delta, \beta_\delta)$.

For the grid-based approximation used by default, choose a rectangular grid

$$\mathcal{G}_n = \left\{(\mu^{(a)}_\delta, \beta^{(b)}_\delta) : a = 1, \ldots, A,\ b = 1, \ldots, B\right\}.$$

At each grid point define

$$W_{ab} = \mathcal{W}_n(\mu^{(a)}_\delta, \beta^{(b)}_\delta), \qquad g_{ab} = g_n(\mu^{(a)}_\delta, \beta^{(b)}_\delta).$$

The grid estimator of the projected interval is

$$\widehat{\mathrm{CI}}^{\Delta,\mathrm{proj,grid}}_{1-\alpha} = \left[\min\{g_{ab} : W_{ab} \le c_{2,\alpha}\}, \max\{g_{ab} : W_{ab} \le c_{2,\alpha}\}\right],$$

provided that the accepted grid set is non-empty. In the software, the grid bounds are enlarged iteratively until the accepted set no longer touches the boundary of the evaluated box, or until a prespecified maximum number of expansions is reached.

For the optimization-based version, let $B_n \subset \mathbb{R}^2$ be the current rectangular search box on the $(\mu_\delta, \beta_\delta)$ scale. The exact endpoint problems are the constrained programs

$$L^{\mathrm{proj}}_{1-\alpha}(B_n) = \inf_{(\mu_\delta,\beta_\delta)\in B_n}\left\{g_n(\mu_\delta, \beta_\delta) : \mathcal{W}_n(\mu_\delta, \beta_\delta) \le c_{2,\alpha}\right\}, \tag{33}$$

$$U_{1-\alpha}^{\text{proj}}(B_n) = \sup_{(\mu_\delta,\beta_\delta)\in B_n} \{g_n(\mu_\delta,\beta_\delta) : \mathcal{W}_n(\mu_\delta,\beta_\delta) \le c_{2,\alpha}\}. \tag{34}$$

The implementation solves these box-constrained problems numerically by replacing the hard inequality with a one-sided quadratic penalty. Writing $x_+ = \max(x, 0)$ and letting $\kappa_n > 0$ denote the large data-scaled penalty constant used in the code, the exact objectives passed to the optimizer are

$$Q_{n,\alpha}^{\text{proj},L}(\mu_\delta,\beta_\delta) = g_n(\mu_\delta,\beta_\delta) + \kappa_n\big(\mathcal{W}_n(\mu_\delta,\beta_\delta) - c_{2,\alpha}\big)_+^2, \tag{35}$$

$$Q_{n,\alpha}^{\text{proj},U}(\mu_\delta,\beta_\delta) = -g_n(\mu_\delta,\beta_\delta) + \kappa_n\big(\mathcal{W}_n(\mu_\delta,\beta_\delta) - c_{2,\alpha}\big)_+^2. \tag{36}$$

These expressions are obtained directly from Equations (33)–(34) by taking the lower-endpoint target $g_n(\mu_\delta,\beta_\delta)$ or the upper-endpoint target $-g_n(\mu_\delta,\beta_\delta)$ and then adding a quadratic penalty for positive excess above the simultaneous-region threshold. After numerical optimization of Equation (35) or (36) within B_n, the software moves along the line segment from the fitted center point to the optimizer proposal and performs binary search on that segment to recover the last feasible point satisfying $\mathcal{W}_n \le c_{2,\alpha}$. If either endpoint lies too close to the box boundary, the box is expanded and the search is repeated.

Profile interval

The Δ-profile statistic is the one-dimensional profile of the same joint surface under the nonlinear constraint in Equation (30). For any target value $\Delta \in \mathbb{R}$ on the atom-at-zero numerical combined-outcome scale, define

$$T_n(\Delta) = \inf\{\mathcal{W}_n(\mu_\delta,\beta_\delta) : g_n(\mu_\delta,\beta_\delta) = \Delta\}. \tag{37}$$

The corresponding profile interval is

$$\text{CI}_{1-\alpha}^{\Delta,\text{prof}} = \{\Delta \in \mathbb{R} : T_n(\Delta) \le c_{1,\alpha}\}.$$

The package reports this set through its lower and upper endpoints,

$$\text{CI}_{1-\alpha}^{\Delta,\text{prof}} = \left[L_{1-\alpha}^{\text{prof}}, U_{1-\alpha}^{\text{prof}}\right],$$

where

$$L_{1-\alpha}^{\text{prof}} = \inf\{\Delta : T_n(\Delta) \le c_{1,\alpha}\}, \qquad U_{1-\alpha}^{\text{prof}} = \sup\{\Delta : T_n(\Delta) \le c_{1,\alpha}\}.$$

Whenever the infimum in Equation (37) is attained on each non-empty fiber $\{(\mu_\delta,\beta_\delta) : g_n(\mu_\delta,\beta_\delta) = \Delta\}$, the accepted Δ values can equivalently be written as

$$\{g_n(\mu_\delta,\beta_\delta) : \mathcal{W}_n(\mu_\delta,\beta_\delta) \le c_{1,\alpha}\},$$

which explains the grid approximation used in the package. Specifically, on the same joint grid $\mathcal{G}_n$ defined above, the default grid estimator is

$$\widehat{\text{CI}}_{1-\alpha}^{\Delta,\text{prof,grid}} = [\min\{g_{ab} : W_{ab} \le c_{1,\alpha}\}, \max\{g_{ab} : W_{ab} \le c_{1,\alpha}\}].$$

Thus the profile and projected grid intervals are computed from the same evaluated joint surface and differ only in the threshold: $c_{1,\alpha}$ for the profile interval and $c_{2,\alpha}$ for the projected interval.

For the optimization-based profile interval, the inner profile problem in Equation (37) is reduced to a one-dimensional optimization over β_δ.

In the parametric Gaussian case, write

$$N_+ = \sum_{i=1}^{n} A_i, \qquad \overline{Y}_+ = \frac{1}{N_+} \sum_{i:A_i=1} Y_i, \qquad \overline{R}_+ = \frac{1}{N_+} \sum_{i:A_i=1} R_i.$$

For fixed μ_δ, the constrained Gaussian fit among observed outcomes minimizes

$$S_n(\mu_0; \mu_\delta) = \sum_{i:A_i=1} \left(Y_i - \mu_0 - \mu_\delta R_i\right)^2.$$

Differentiating with respect to μ_0 and setting the derivative to zero gives

$$\widehat{\mu}_0(\mu_\delta) = \overline{Y}_+ - \overline{R}_+ \mu_\delta, \qquad \widehat{\mu}_1(\mu_\delta) = \overline{Y}_+ + (1 - \overline{R}_+)\mu_\delta.$$

Substituting these expressions into Equation (30) yields the exact affine representation on the atom-at-zero numerical combined-outcome scale

$$\begin{aligned} g_n(\mu_\delta, \beta_\delta) = {} & \big(\widehat{\pi}_1(\beta_\delta) - \widehat{\pi}_0(\beta_\delta)\big)\overline{Y}_+ \\ & + \big\{\widehat{\pi}_1(\beta_\delta)(1 - \overline{R}_+) + \widehat{\pi}_0(\beta_\delta)\overline{R}_+\big\}\mu_\delta. \end{aligned} \tag{38}$$

Therefore, for fixed target Δ and fixed β_δ, the unique compatible mean-difference value is

$$\mu_\delta^\star(\Delta, \beta_\delta) = \frac{\Delta - \big(\widehat{\pi}_1(\beta_\delta) - \widehat{\pi}_0(\beta_\delta)\big)\overline{Y}_+}{\widehat{\pi}_1(\beta_\delta)(1 - \overline{R}_+) + \widehat{\pi}_0(\beta_\delta)\overline{R}_+}, \tag{39}$$

Whenever the denominator is positive, Equation (37) reduces exactly to the one-dimensional problem

$$T_n(\Delta) = \inf_{\beta_\delta \in \mathbb{R}} Q_n^{\mathrm{prof,par}}(\Delta, \beta_\delta), \qquad Q_n^{\mathrm{prof,par}}(\Delta, \beta_\delta) = \mathcal{W}_n\big(\mu_\delta^\star(\Delta, \beta_\delta), \beta_\delta\big). \tag{40}$$

In the semi-parametric case, the same reduction is available conceptually but not in closed form. For fixed μ_δ, let

$$\widehat{\mu}_0(\mu_\delta) \in \arg\min_{\mu_0 \in \mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, \mu_\delta), \qquad \widehat{\mu}_1(\mu_\delta) = \widehat{\mu}_0(\mu_\delta) + \mu_\delta,$$

so that Equation (30) defines a nonlinear map $(\mu_\delta, \beta_\delta) \mapsto g_n(\mu_\delta, \beta_\delta)$. For fixed (Δ, β_δ), the compatible mean-difference value $\mu_\delta^\star(\Delta, \beta_\delta)$ is any feasible solution of

$$g_n(\mu_\delta, \beta_\delta) = \Delta.$$

Hence the exact profile objective is

$$Q_n^{\mathrm{prof,sp}}(\Delta, \beta_\delta) = \begin{cases} \mathcal{W}_n\big(\mu_\delta^\star(\Delta, \beta_\delta), \beta_\delta\big), & \text{if a feasible solution exists,} \\ \infty, & \text{otherwise.} \end{cases} \tag{41}$$

The implementation constructs $\mu_\delta^\star(\Delta, \beta_\delta)$ by searching over the empirical-likelihood feasible range

$$\left[\min_{j \in \mathcal{I}_1} Y_j - \max_{i \in \mathcal{I}_0} Y_i,\ \max_{j \in \mathcal{I}_1} Y_j - \min_{i \in \mathcal{I}_0} Y_i\right].$$

If the function $\mu_\delta \mapsto g_n(\mu_\delta, \beta_\delta) - \Delta$ changes sign over that interval, the code solves the equation by root-finding. Otherwise it minimizes the exact squared-mismatch objective

$$M_n(\mu_\delta; \Delta, \beta_\delta) = \{g_n(\mu_\delta, \beta_\delta) - \Delta\}^2 \tag{42}$$

over the feasible interval and accepts the resulting candidate only if the remaining discrepancy is below numerical tolerance. The outer profile statistic is then

$$T_n(\Delta) = \inf_{\beta_\delta \in \mathbb{R}} Q_n^{\mathrm{prof,sp}}(\Delta, \beta_\delta). \tag{43}$$

Once the profile function $\Delta \mapsto T_n(\Delta)$ has been defined, the lower and upper profile endpoints are the roots of $T_n(\Delta) = c_{1,\alpha}$ on the two sides of the fitted value $\widehat{\Delta}_n$. Numerically, the software brackets these roots by step-doubling away from $\widehat{\Delta}_n$ and then refines the bracket by bisection until the desired tolerance is reached.

C Supplementary derivation of the semi-parametric test statistic

This appendix records the constrained empirical-likelihood calculation underlying Equation (14). The grouped expression derived below is algebraically equivalent to the full-sample representation in Equation (14) because $\widetilde{Y}_i^{\,0} = 0$ whenever $A_i = 0$. Throughout, the treatment-group means among the observed outcomes are parameterized as μ_0 for group $R = 0$ and $\mu_1 = \mu_0 + \mu_\delta$ for group $R = 1$.

More explicitly, the full-sample notation in Equation (14) uses the factors $1 - R_i$ and R_i to select the control and treatment contributions, while the factors involving A_i suppress subjects with $A_i = 0$. If $R_i = 0$ and $A_i = 1$, then $\widetilde{Y}_i^{\,0} = Y_i$ and the control-arm summand reduces to $\log\{1 + \lambda_0(Y_i - \mu_0)\}$. If $R_i = 1$ and $A_i = 1$, then the treatment-arm summand reduces to $\log\{1 + \lambda_1(Y_i - \mu_0 - \mu_\delta)\}$. If $A_i = 0$, then $\widetilde{Y}_i^{\,0} = 0$ and both summands equal $\log(1) = 0$. Thus the full-sample display is exactly the same criterion as the grouped notation indexed by $\mathcal{I}_0$ and $\mathcal{I}_1$.

Lemma 4. *Fix $(\mu_0, \mu_\delta) \in \mathbb{R}^2$ such that μ_0 lies in the interior of the convex hull of $\{Y_i : i \in \mathcal{I}_0\}$ and $\mu_0 + \mu_\delta$ lies in the interior of the convex hull of $\{Y_j : j \in \mathcal{I}_1\}$. Define the normalized empirical likelihood ratio*

$$R_{Y,n}^{\mathrm{E}}(\mu_0, \mu_\delta) = \sup_{\{p_i\},\{q_j\}} \left\{ \prod_{i \in \mathcal{I}_0} |\mathcal{I}_0| p_i \prod_{j \in \mathcal{I}_1} |\mathcal{I}_1| q_j \right\} \tag{44}$$

subject to

$$\begin{aligned} p_i \geq 0, \quad & \sum_{i \in \mathcal{I}_0} p_i = 1, \quad & \sum_{i \in \mathcal{I}_0} p_i (Y_i - \mu_0) = 0, \\ q_j \geq 0, \quad & \sum_{j \in \mathcal{I}_1} q_j = 1, \quad & \sum_{j \in \mathcal{I}_1} q_j (Y_j - \mu_0 - \mu_\delta) = 0. \end{aligned}$$

Then the maximizer in Equation (44) is unique and is given by

$$p_i = \frac{1}{|\mathcal{I}_0| \left(1 + \lambda_0 (Y_i - \mu_0)\right)}, \qquad q_j = \frac{1}{|\mathcal{I}_1| \left(1 + \lambda_1 (Y_j - \mu_0 - \mu_\delta)\right)},$$

where (λ_0, λ_1) *satisfy*

$$\frac{1}{|\mathcal{I}_0|}\sum_{i\in\mathcal{I}_0}\frac{Y_i-\mu_0}{1+\lambda_0(Y_i-\mu_0)}=0,$$
$$\frac{1}{|\mathcal{I}_1|}\sum_{j\in\mathcal{I}_1}\frac{Y_j-\mu_0-\mu_\delta}{1+\lambda_1(Y_j-\mu_0-\mu_\delta)}=0.$$

Consequently,

$$\begin{aligned}\log R^{\mathrm{E}}_{Y,n}(\mu_0,\mu_\delta) &= -\sum_{i\in\mathcal{I}_0}\log\left(1+\lambda_0(Y_i-\mu_0)\right)-\sum_{j\in\mathcal{I}_1}\log\left(1+\lambda_1(Y_j-\mu_0-\mu_\delta)\right),\\ \widetilde{W}_{Y,n}(\mu_0,\mu_\delta) &= -2\log R^{\mathrm{E}}_{Y,n}(\mu_0,\mu_\delta)\\ &= 2\left(\sum_{i\in\mathcal{I}_0}\log\left(1+\lambda_0(Y_i-\mu_0)\right)+\sum_{j\in\mathcal{I}_1}\log\left(1+\lambda_1(Y_j-\mu_0-\mu_\delta)\right)\right),\end{aligned}$$

and the profiled statistic is

$$\widetilde{W}^{P}_{Y,n}(\mu_\delta)=\inf_{\mu_0\in\mathbb{R}}\widetilde{W}_{Y,n}(\mu_0,\mu_\delta).$$

Proof. The objective in Equation (44) is strictly concave on a convex feasible set. The convex-hull assumptions imply that the feasible set contains an interior point, hence the maximizer is unique and satisfies the first-order conditions. Introduce the Lagrangian

$$\begin{aligned}O(\{p_i\},\{q_j\}) &= \sum_{i\in\mathcal{I}_0}\log(|\mathcal{I}_0|p_i)+\sum_{j\in\mathcal{I}_1}\log(|\mathcal{I}_1|q_j)\\ &\quad+\gamma_1\left(\sum_{i\in\mathcal{I}_0}p_i-1\right)+\gamma_2\left(\sum_{j\in\mathcal{I}_1}q_j-1\right)\\ &\quad-|\mathcal{I}_0|\lambda_0\sum_{i\in\mathcal{I}_0}p_i(Y_i-\mu_0)-|\mathcal{I}_1|\lambda_1\sum_{j\in\mathcal{I}_1}q_j(Y_j-\mu_0-\mu_\delta).\end{aligned}\tag{45}$$

Differentiation with respect to p_i and q_j yields

$$0=\frac{\partial O(\{p_i\},\{q_j\})}{\partial p_i}=\frac{1}{p_i}-|\mathcal{I}_0|\lambda_0(Y_i-\mu_0)+\gamma_1,\tag{46}$$
$$0=\frac{\partial O(\{p_i\},\{q_j\})}{\partial q_j}=\frac{1}{q_j}-|\mathcal{I}_1|\lambda_1(Y_j-\mu_0-\mu_\delta)+\gamma_2.\tag{47}$$

Multiplying Equation (46) by p_i and summing over $i\in\mathcal{I}_0$, then using the constraints, gives

$$\begin{aligned}0 &= \sum_{i\in\mathcal{I}_0}p_i\frac{\partial O(\{p_i\},\{q_j\})}{\partial p_i}\\ &= |\mathcal{I}_0|-|\mathcal{I}_0|\lambda_0\sum_{i\in\mathcal{I}_0}p_i(Y_i-\mu_0)+\gamma_1\sum_{i\in\mathcal{I}_0}p_i\\ &= |\mathcal{I}_0|+\gamma_1.\end{aligned}$$

Hence $\gamma_1=-|\mathcal{I}_0|$, and the same argument gives $\gamma_2=-|\mathcal{I}_1|$. Substituting these identities into Equations (46) and (47) yields the stated optimizer weights.

The multiplier equations follow by imposing the moment constraints after substitution of the optimizer. Inserting the optimizer into Equation (44) gives

$$\begin{aligned}\log R^{\mathrm{E}}_{Y,n}(\mu_0,\mu_\delta) &= \sum_{i\in\mathcal{I}_0}\log(|\mathcal{I}_0|p_i) + \sum_{j\in\mathcal{I}_1}\log(|\mathcal{I}_1|q_j)\\ &= -\sum_{i\in\mathcal{I}_0}\log\left(1+\lambda_0(Y_i-\mu_0)\right) - \sum_{j\in\mathcal{I}_1}\log\left(1+\lambda_1(Y_j-\mu_0-\mu_\delta)\right).\end{aligned}$$

The displayed expression for $\widetilde{W}_{Y,n}(\mu_0,\mu_\delta)$ is therefore immediate from the definition $\widetilde{W}_{Y,n}(\mu_0,\mu_\delta) = -2\log R^{\mathrm{E}}_{Y,n}(\mu_0,\mu_\delta)$, and the profiled statistic is obtained by minimizing over the nuisance mean μ_0. □

References

John Aitchison. On the distribution of a positive random variable having a discrete probability mass at the origin. *Journal of the american statistical association*, 50(271):901 – 908, 1955.

Charles E. Antoniak. Mixtures of dirichlet processes with applications to bayesian nonparametric problems. *The Annals of Statistics*, 2(6):1152–1174, 1974.

Hugh A. Chipman, Edward I. George, and Robert E. McCulloch. BART: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1):266–298, 2010. doi: 10.1214/09-AOAS285.

Elizabeth Colantuoni, Daniel O Scharfstein, Chenguang Wang, Mohamed D Hashem, Andrew Leroux, Dale M Needham, and Timothy D Girard. Statistical methods to compare functional outcomes in randomized controlled trials with high mortality. *BMJ*, 360:j5748, 2018. doi: 10.1136/bmj.j5748.

John G Cragg. Some statistical models for limited dependent variables with application to the demand for durable goods. *Econometrica: Journal of the Econometric Society*, pages 829–844, 1971.

Maria De Iorio, Peter Müller, Gary L. Rosner, and Steven N. MacEachern. An ANOVA model for dependent random measures. *Journal of the American Statistical Association*, 99(465):205–215, 2004. doi: 10.1198/016214504000000205.

Thomas S. Ferguson. A bayesian analysis of some nonparametric problems. *The Annals of Statistics*, 1(2):209–230, 1973.

Constantine E Frangakis and Donald B Rubin. Principal stratification in causal inference. *Biometrics*, 58(1):21–29, 2002. doi: 10.1111/j.0006-341X.2002.00021.x.

Alan E. Gelfand, Athanasios Kottas, and Steven N. MacEachern. Bayesian nonparametric spatial modeling with dirichlet process mixing. *Journal of the American Statistical Association*, 100 (471):1021–1035, 2005. doi: 10.1198/016214504000002078.

Andrew Gelman, Xiao-Li Meng, and Hal Stern. Posterior predictive assessment of model fitness via realized discrepancies. *Statistica Sinica*, 6(4):733–807, 1996.

Hemant Ishwaran and Lancelot F. James. Gibbs sampling methods for stick-breaking priors. *Journal of the American Statistical Association*, 96(453):161–173, 2001.

IST-3 Collaborative Group. The benefits and harms of intravenous thrombolysis with recombinant tissue plasminogen activator within 6 h of acute ischaemic stroke (the third international stroke trial [IST-3]): A randomised controlled trial. *The Lancet*, 379(9834):2352–2363, 2012. doi: 10.1016/S0140-6736(12)60768-5.

Andreas Kryger Jensen and Theis Lange. The TruncComp R package for two-sample comparison of truncated continuous outcomes using parametric and semi-empirical likelihood. https://github.com/aejensen/TruncComp, 2018.

Yuichi Kitamura. Asymptotic optimality of empirical likelihood for testing moment restrictions. *Econometrica*, 69(6):1661–1672, 2001.

John M Lachin. Worst-rank score analysis with informatively missing observations in clinical trials. *Controlled Clinical Trials*, 20(5):408–422, 1999. doi: 10.1016/S0197-2456(99)00022-7.

Steven N. MacEachern. Dependent nonparametric processes. In *ASA Proceedings of the Section on Bayesian Statistical Science*, pages 50–55, Alexandria, VA, 1999. American Statistical Association.

Henry B Mann and Donald R Whitney. On a test of whether one of two random variables is stochastically larger than the other. *The annals of mathematical statistics*, pages 50–60, 1947.

Robert P McMahon and Frank E Harrell. Power calculation for clinical trials when the outcome is a composite ranking of survival and a nonfatal outcome. *Controlled Clinical Trials*, 21(4): 305–312, 2000. doi: 10.1016/S0197-2456(00)00052-0.

Art Owen. Empirical likelihood ratio confidence intervals for a single functional. *Biometrika*, 75 (2):237–249, 1988.

Art Owen. Empirical likelihood for linear models. *The Annals of Statistics*, pages 1725–1747, 1991.

Jing Qin and Jerry Lawless. Empirical likelihood and general estimating equations. *The Annals of Statistics*, 22(1):300–325, 1994. doi: 10.1214/aos/1176325370.

Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA, 2006.

Peter Sandercock, Joanna Wardlaw, Richard Lindley, Geoff Cohen, and William Whiteley. The third international stroke trial (IST-3), 2000–2015 [dataset], 2016. URL https://doi.org/10.7488/ds/1350.

Jayaram Sethuraman. A constructive definition of dirichlet priors. *Statistica Sinica*, 4(2):639–650, 1994.

Yee Whye Teh, Michael I. Jordan, Matthew J. Beal, and David M. Blei. Hierarchical dirichlet processes. *Journal of the American Statistical Association*, 101(476):1566–1581, 2006. doi: 10.1198/016214506000000302.

The COVID STEROID 2 Trial Group. Effect of 12 mg vs 6 mg of Dexamethasone on the Number of Days Alive Without Life Support in Adults With COVID-19 and Severe Hypoxemia: The COVID STEROID 2 Randomized Trial. *JAMA*, 326(18):1807–1817, 11 2021. ISSN 0098-7484.

Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, pages 80–83, 1945.

Samuel S Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, 1938.

Nadir Yehya, Michael O Harhay, Martha A. Q. Curley, David A. Schoenfeld, and Ron W. Reeder. Reappraisal of ventilator-free days in critical care research. *American Journal of Respiratory and Critical Care Medicine*, 200(7):828–836, 2019. doi: 10.1164/rccm.201810-2050CP.